KAZAKH-BRITISH TECHNICAL UNIVERSITY

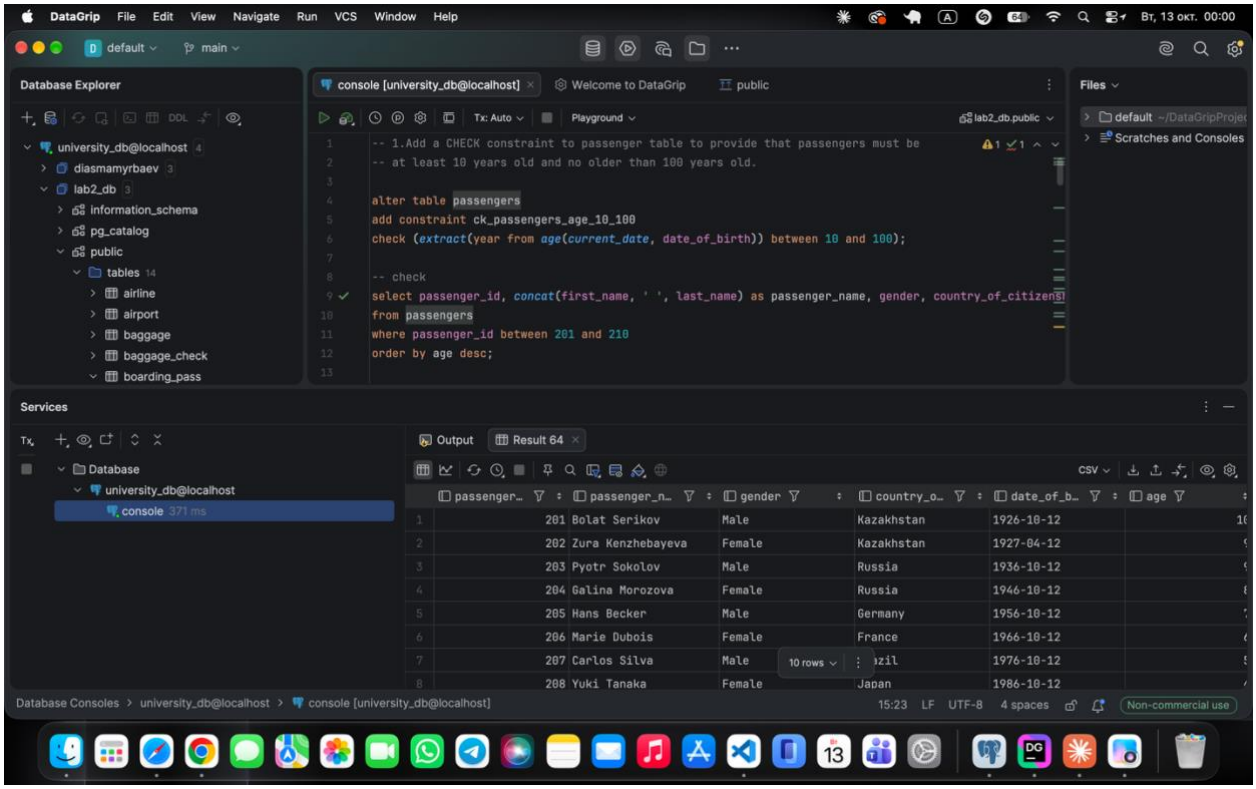
Course name: Databases

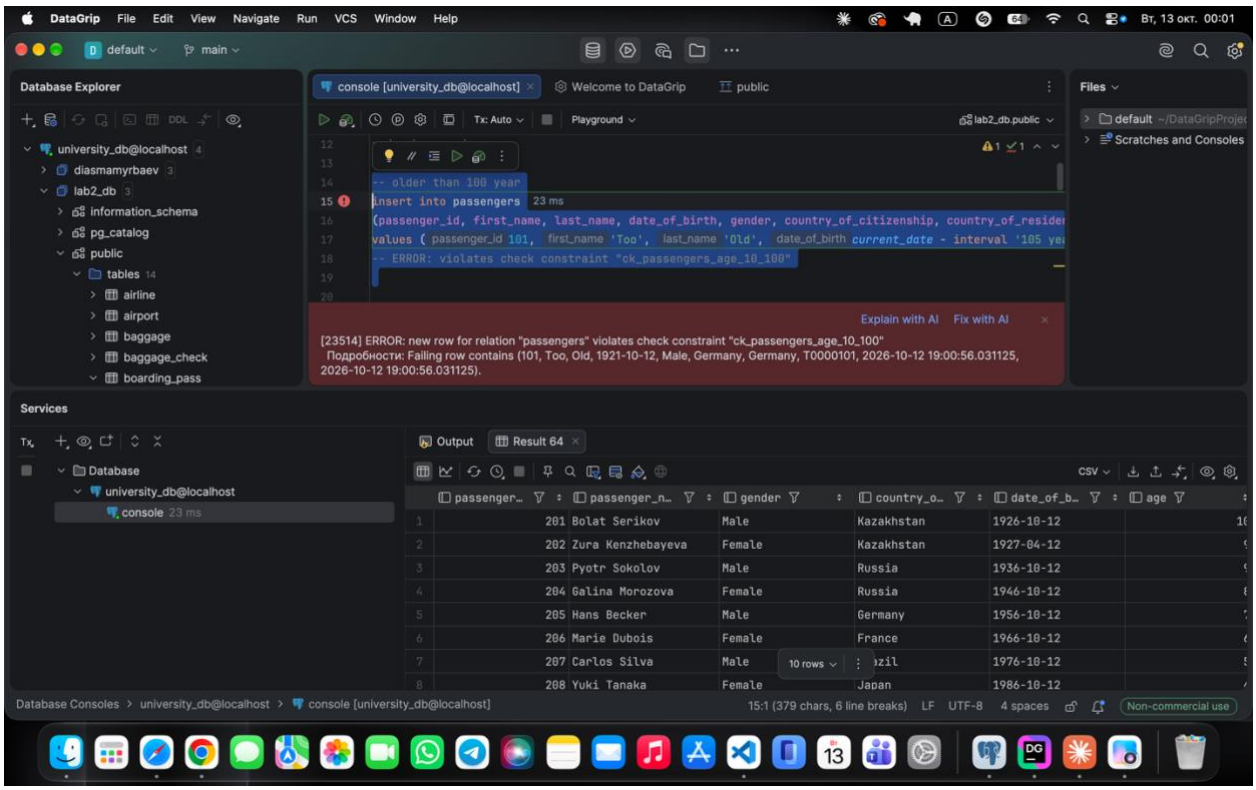
Laboratory work 5

Submission format: please work with the database from the prev lab work and provide full-page screenshots that covers the code and results of each task in the PDF format, also provide sql files.

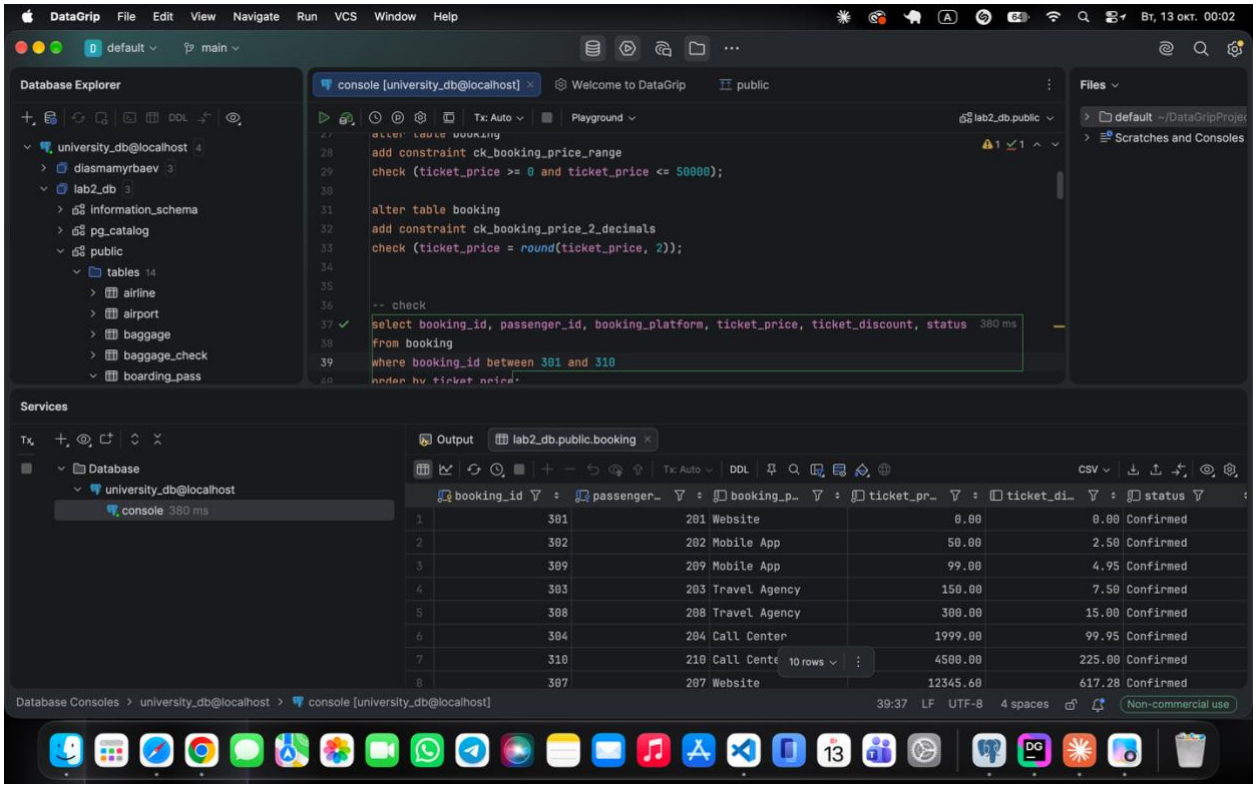
Required tasks:

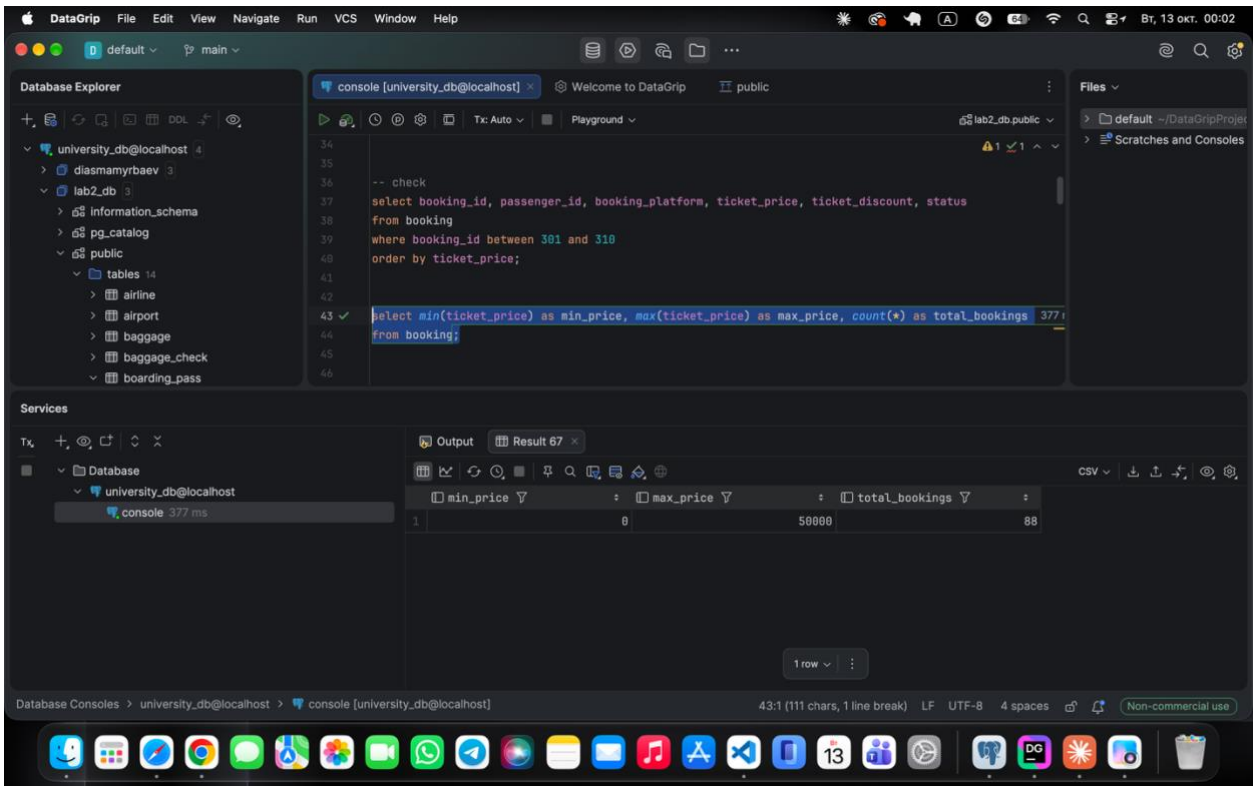
1. Add a CHECK constraint to passenger table to provide that passengers must be at least 10 years old and no older than 100 years old.

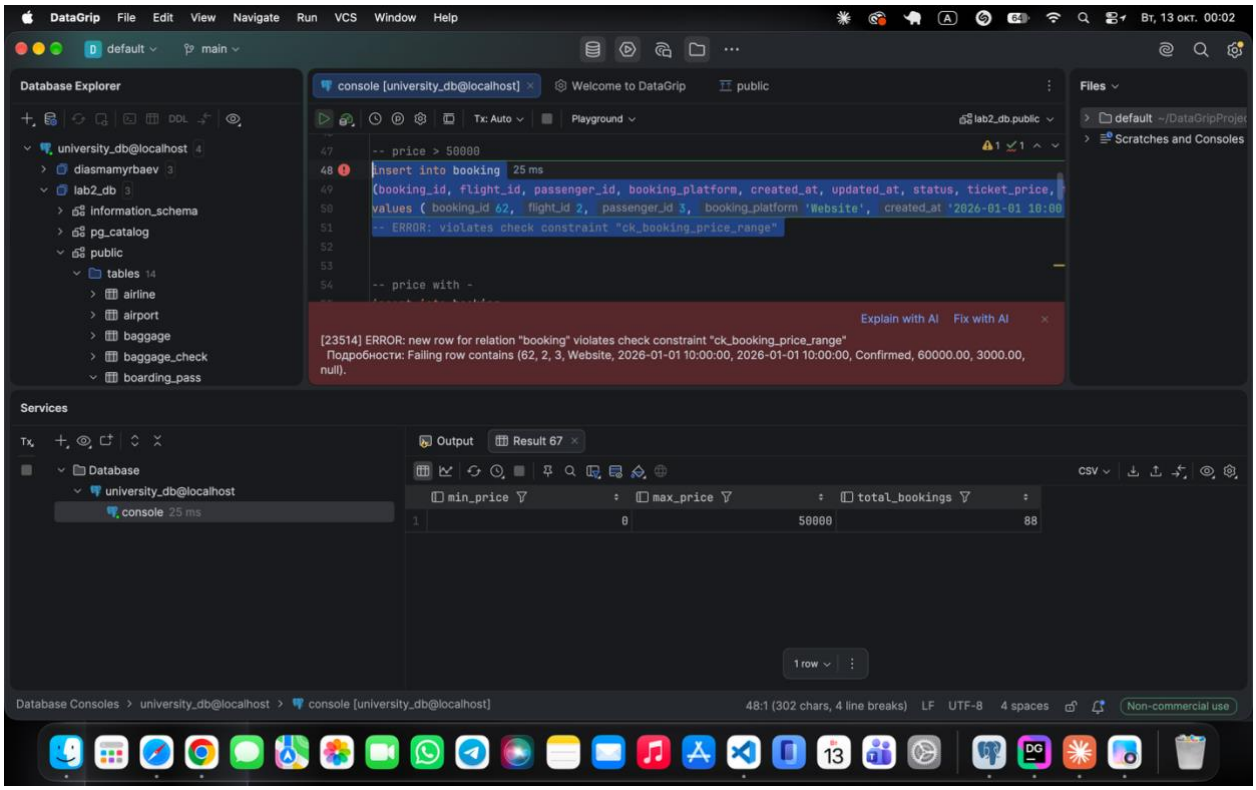


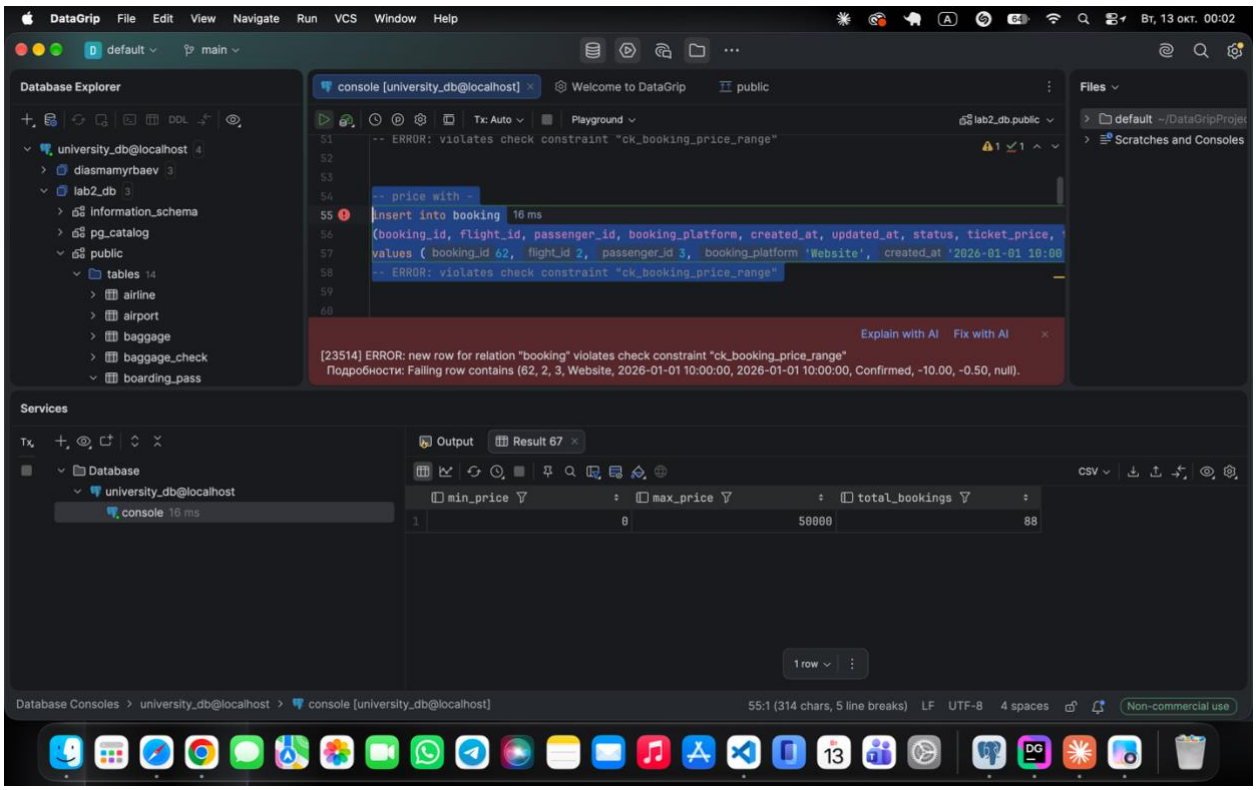


2. Add a CHECK constraint to accept values in booking price not more than 50000tg and less than 0tg. Additionally, the price must be specified with no more than two decimal places.

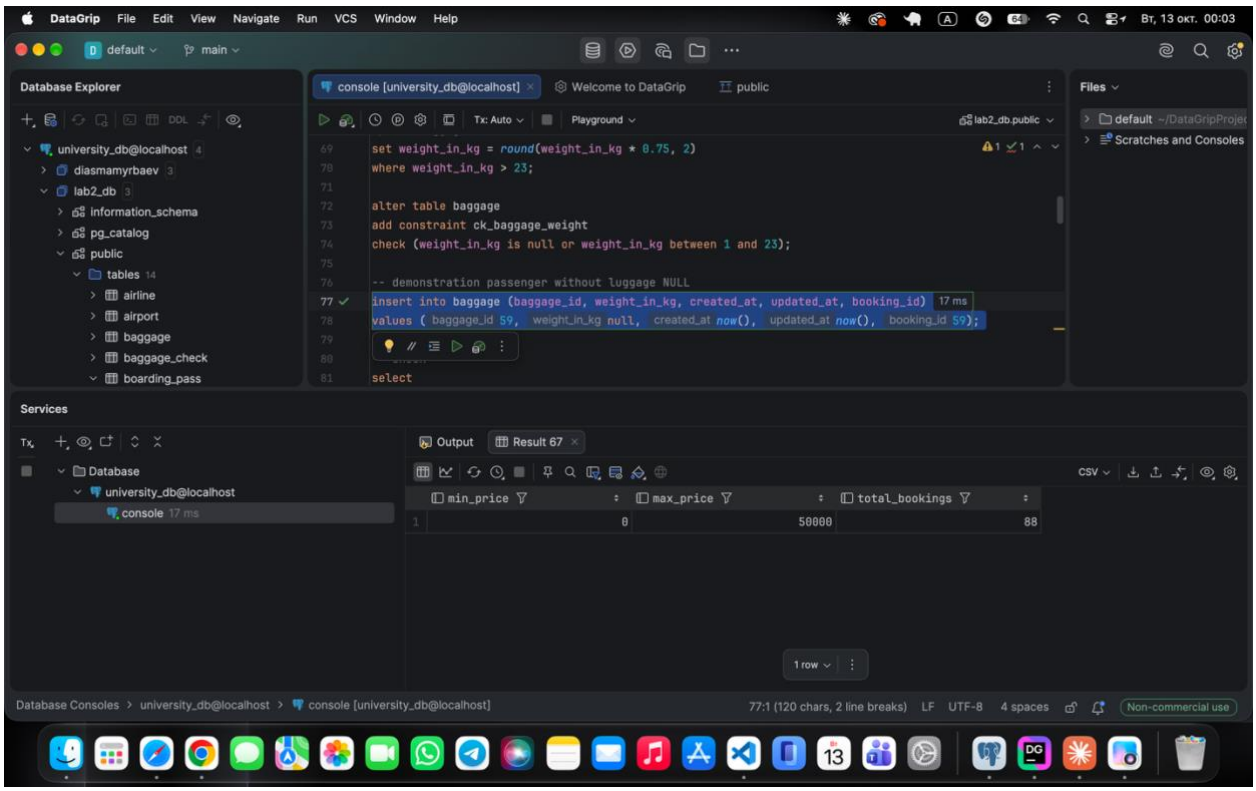


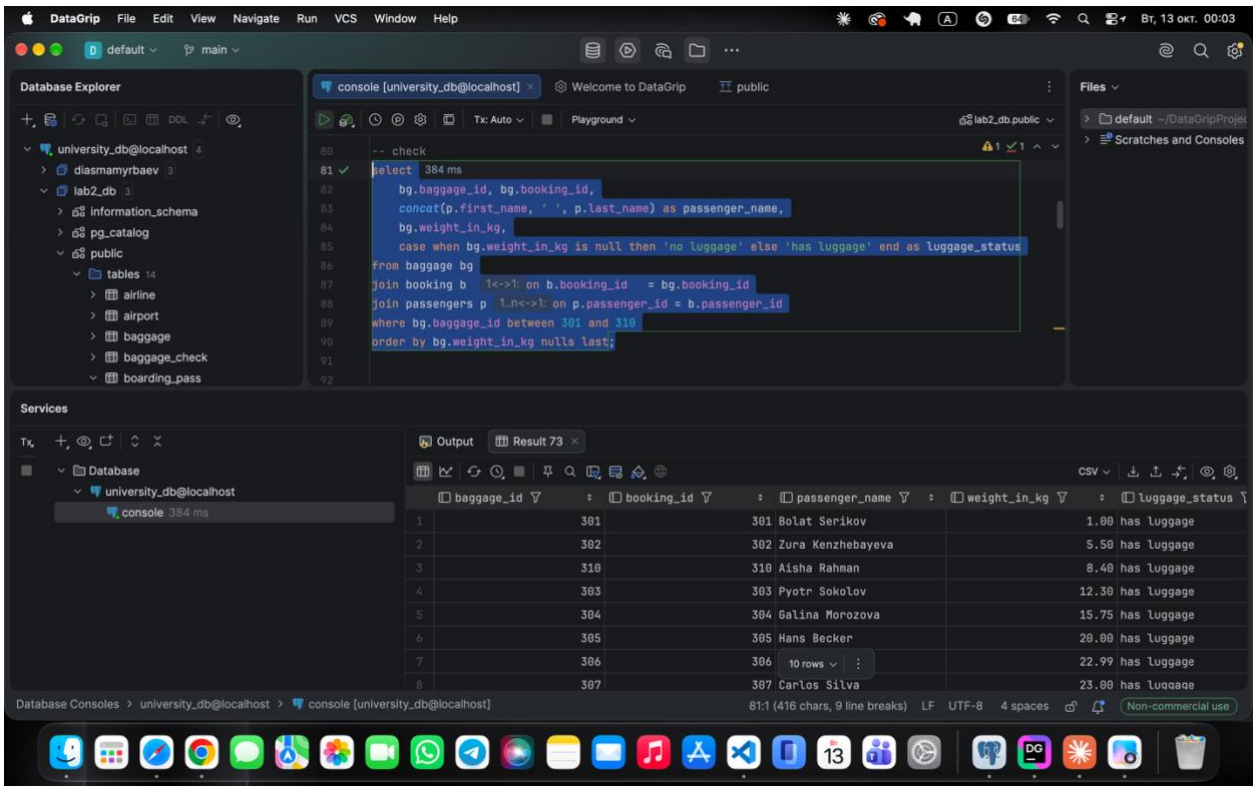

DataGrip File Edit View Navigate Run VCS Window Help
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
-- check
select booking_id, passenger_id, booking_platform, ticket_price, ticket_discount, status
from booking
where booking_id between 301 and 310
order by ticket_price;
select min(ticket_price) as min_price, max(ticket_price) as max_price, count(*) as total_bookings
from booking;
Services
Output
Result 67
min_price
max_price
total_bookings
0
50000
88
Database Consoles > university_db@localhost > console [university_db@localhost]
Non-commercial use

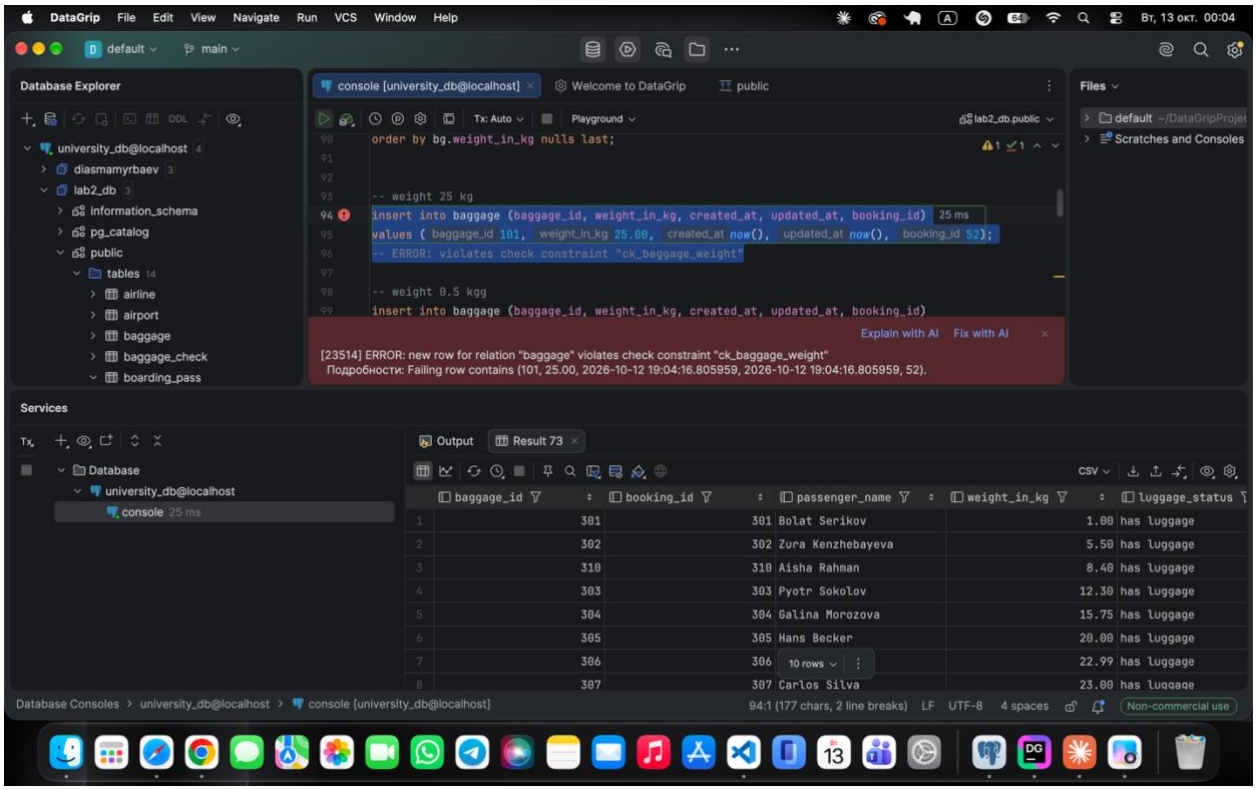

DataGrip File Edit View Navigate Run VCS Window Help
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
-- price > 50000
insert into booking
(booking_id, flight_id, passenger_id, booking_platform, created_at, updated_at, status, ticket_price,
values (booking_id 62, flight_id 2, passenger_id 3, booking_platform 'Website', created_at '2026-01-01 10:00
-- ERROR: violates check constraint "ck_booking_price_range"
-- price with -
Explain with AI
Fix with AI
[23514] ERROR: new row for relation "booking" violates check constraint "ck_booking_price_range"
Подробности: Failing row contains (62, 2, 3, Website, 2026-01-01 10:00:00, 2026-01-01 10:00:00, Confirmed, 60000.00, 3000.00, null).
Services
Output
Result 67
min_price
max_price
total_bookings
0
50000
88
Database Consoles > university_db@localhost > console [university_db@localhost]
Non-commercial use

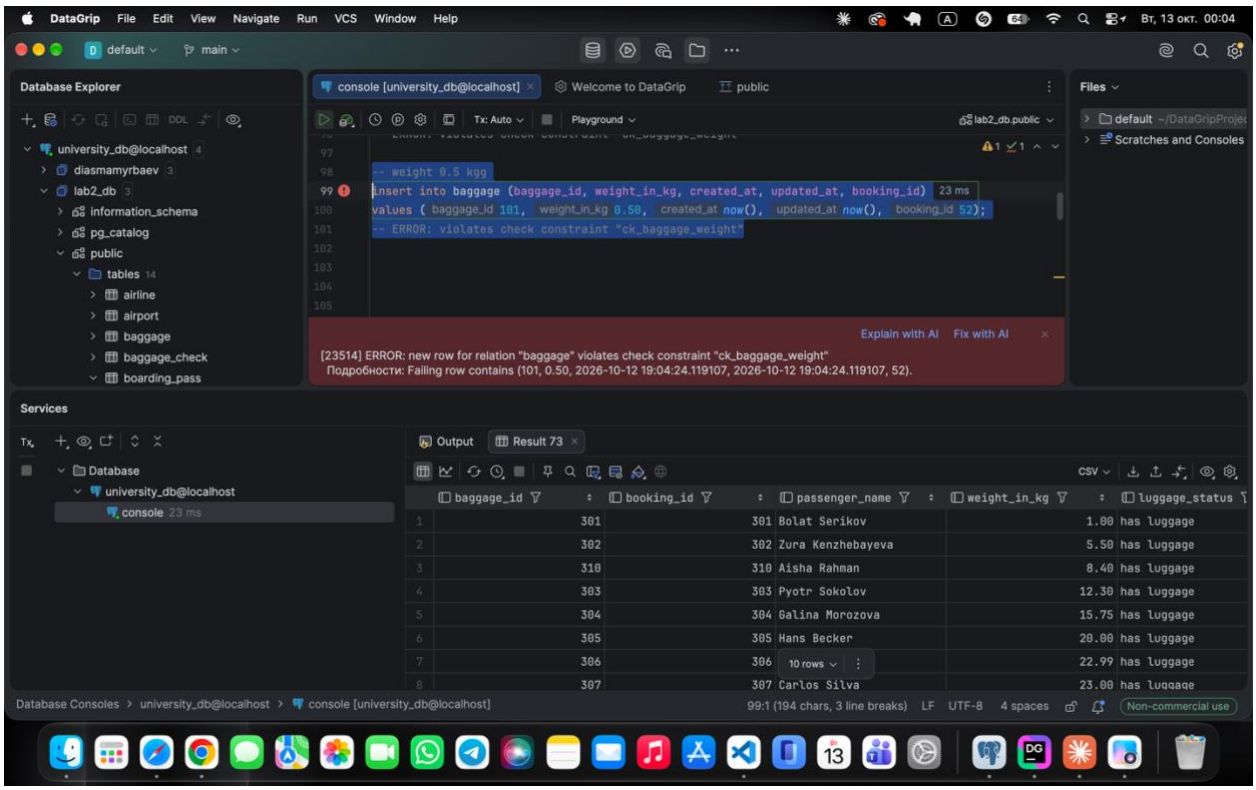


3. Add a CHECK constraint to accept the luggage weight between 1 and 23 kg. If a passenger has no luggage, the value should be allowed to remain NULL.

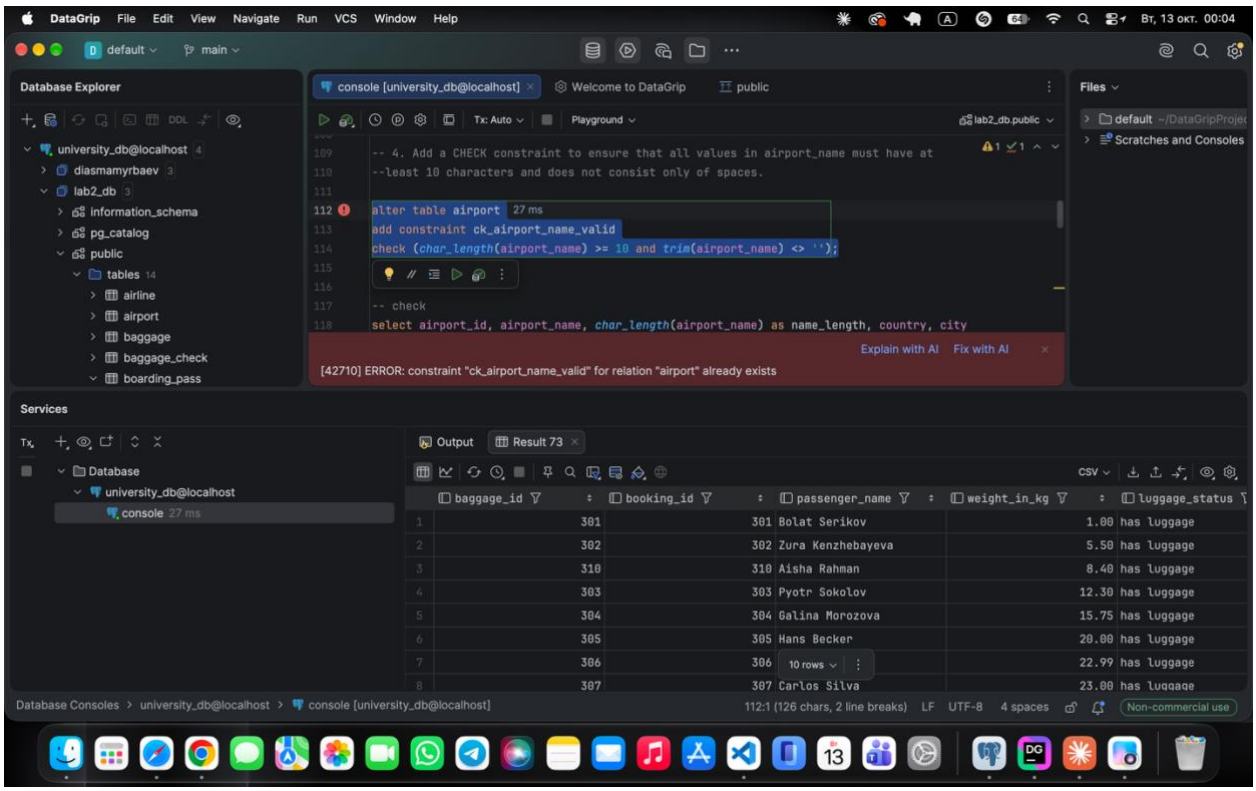


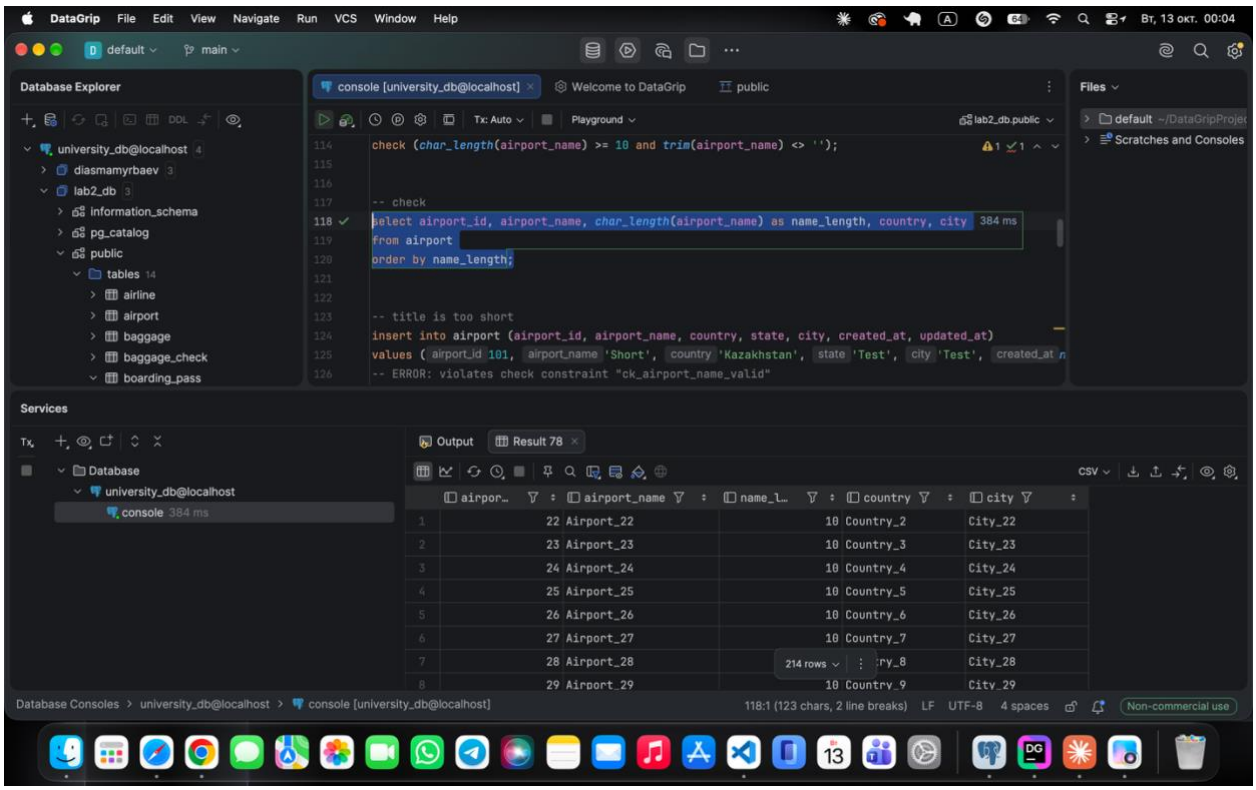

DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:03
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
-- check
select 384 ms
bg.baggage_id, bg.booking_id,
concat(p.first_name, ' ', p.last_name) as passenger_name,
bg.weight_in_kg,
case when bg.weight_in_kg is null then 'no luggage' else 'has luggage' end as luggage_status
from baggage bg
join booking b 1<->1 on b.booking_id = bg.booking_id
join passengers p 1..n<->1 on p.passenger_id = b.passenger_id
where bg.baggage_id between 301 and 310
order by bg.weight_in_kg nulls last;
Services
Output
Result 73
baggage_id booking_id passenger_name weight_in_kg luggage_status
301 301 Bolat Serikov 1.00 has luggage
302 302 Zura Kenzhebayeva 5.50 has luggage
310 310 Aisha Rahman 8.40 has luggage
303 303 Pyotr Sokolov 12.30 has luggage
304 304 Galina Morozova 15.75 has luggage
305 305 Hans Becker 20.00 has luggage
306 306 22.99 has luggage
307 307 Carlos Silva 23.00 has luggage
10 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
81:1 (416 chars, 9 line breaks) LF UTF-8 4 spaces
Non-commercial use

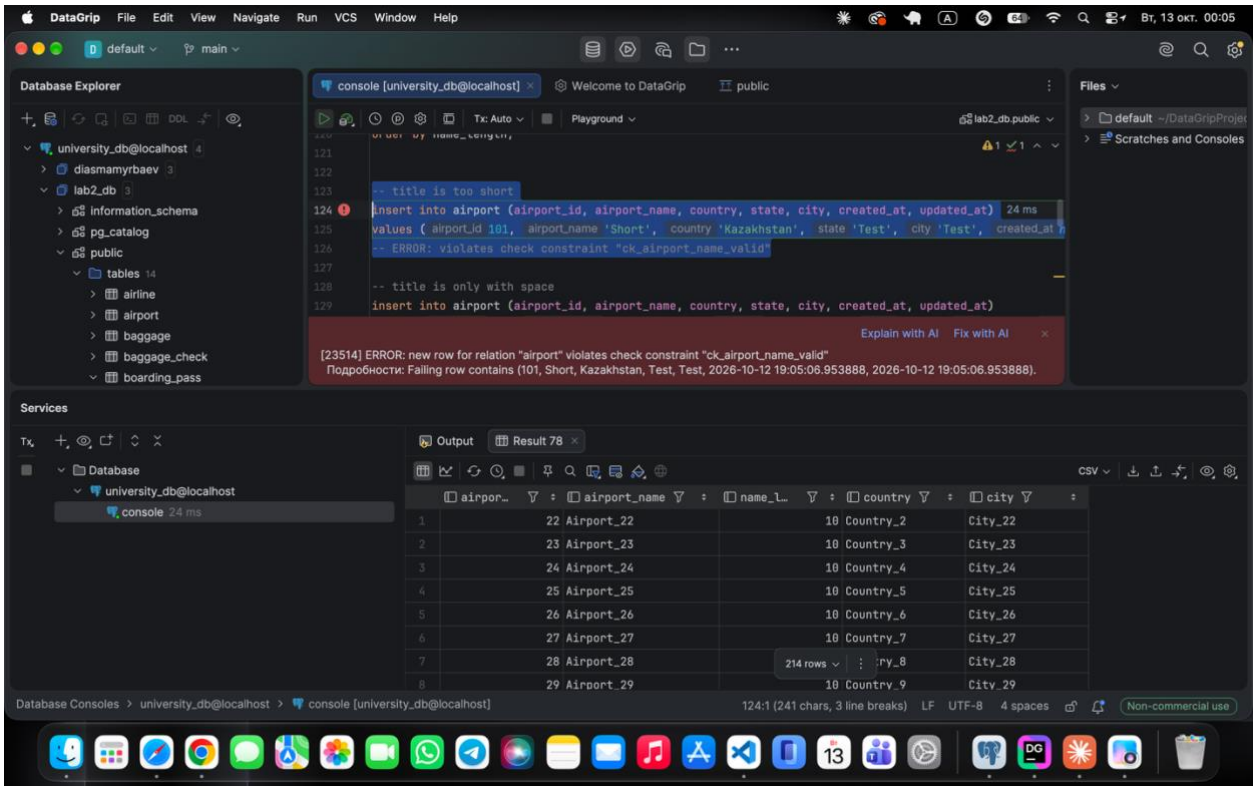

DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:04
order by bg.weight_in_kg nulls last;
-- weight 25 kg
insert into baggage (baggage_id, weight_in_kg, created_at, updated_at, booking_id) 25 ms
values (baggage_id 101, weight_in_kg 25.00, created_at now(), updated_at now(), booking_id 52);
-- ERROR: violates check constraint "ck_baggage_weight"
-- weight 0.5 kgg
insert into baggage (baggage_id, weight_in_kg, created_at, updated_at, booking_id)
Explain with AI Fix with AI
[23514] ERROR: new row for relation "baggage" violates check constraint "ck_baggage_weight"
Подробности: Failing row contains (101, 25.00, 2026-10-12 19:04:16.805959, 2026-10-12 19:04:16.805959, 52).
console 25 ms
94:1 (177 chars, 2 line breaks) LF UTF-8 4 spaces
Non-commercial use

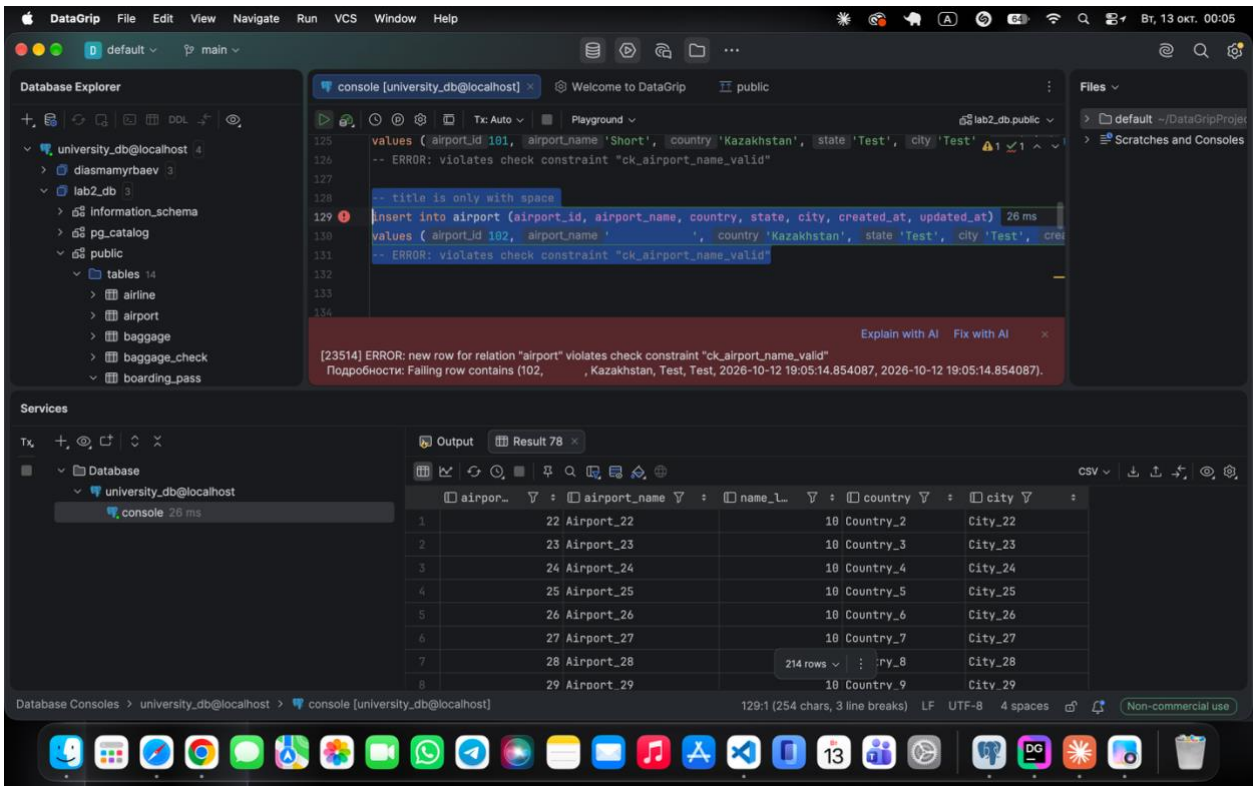


4. Add a CHECK constraint to ensure that all values in airport_name must have at least 10 characters and does not consist only of spaces.

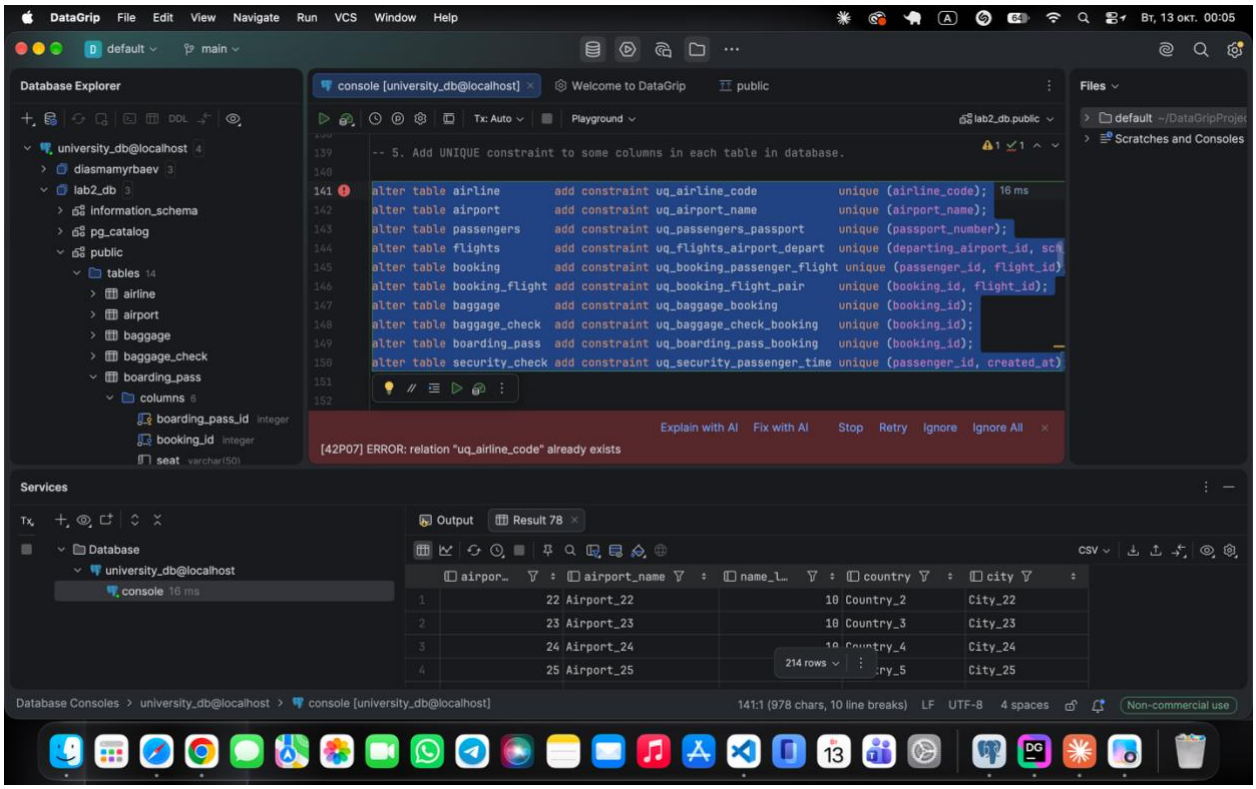


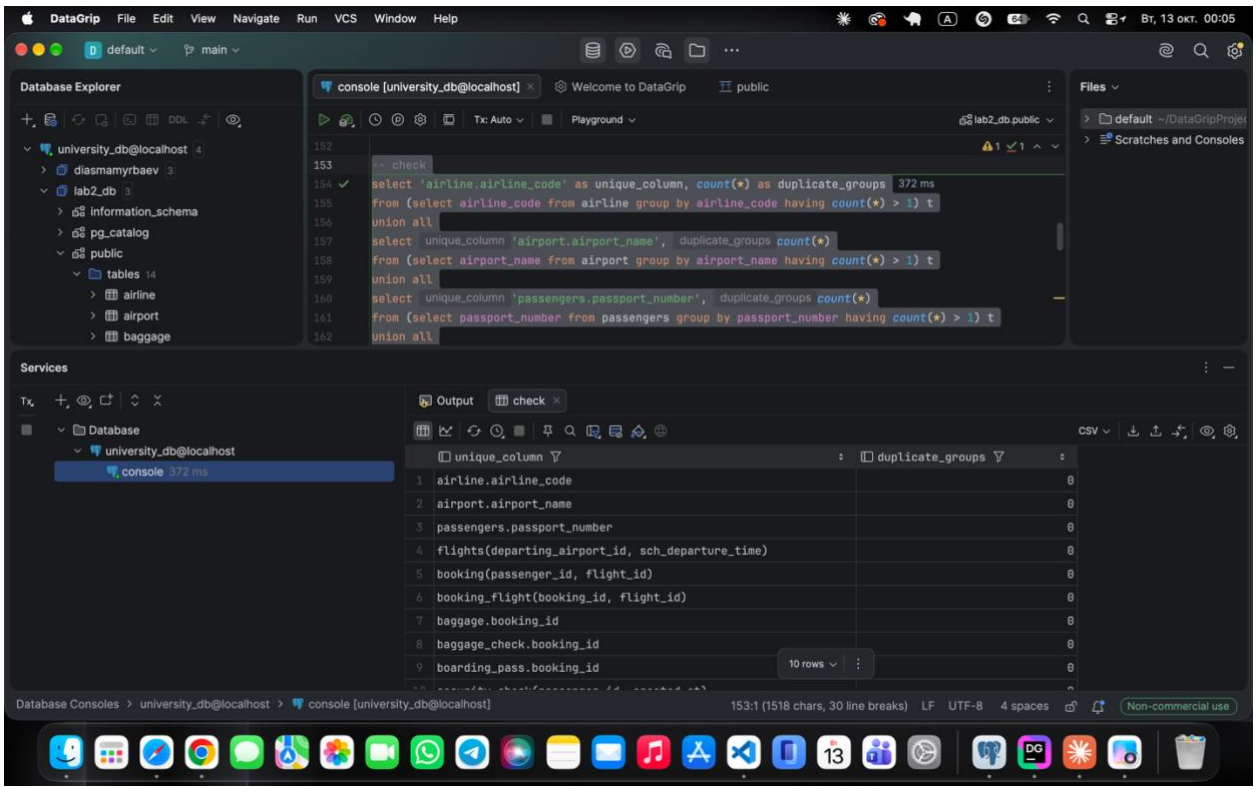

DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:04
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
check (char_length(airport_name) >= 10 and trim(airport_name) <> '');
-- check
select airport_id, airport_name, char_length(airport_name) as name_length, country, city
384 ms
from airport
order by name_length;
-- title is too short
insert into airport (airport_id, airport_name, country, state, city, created_at, updated_at)
values (airport_id 101, airport_name 'Short', country 'Kazakhstan', state 'Test', city 'Test',
-- ERROR: violates check constraint "ck_airport_name_valid"
Services
Output
Result 78
214 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
118:1 (123 chars, 2 line breaks) LF UTF-8 4 spaces
Non-commercial use

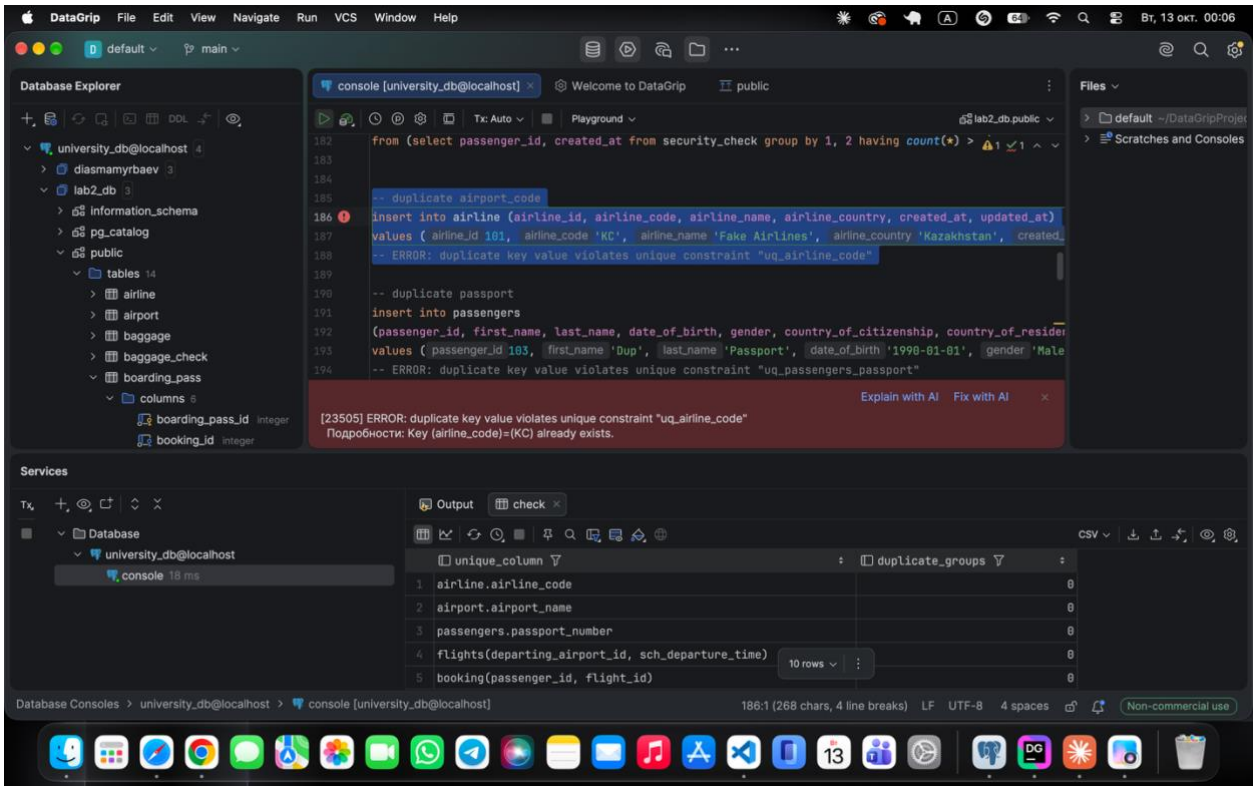

DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:05
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
-- title is too short
insert into airport (airport_id, airport_name, country, state, city, created_at, updated_at)
24 ms
values (airport_id 101, airport_name 'Short', country 'Kazakhstan', state 'Test', city 'Test',
-- ERROR: violates check constraint "ck_airport_name_valid"
-- title is only with space
insert into airport (airport_id, airport_name, country, state, city, created_at, updated_at)
Explain with AI
Fix with AI
[23514] ERROR: new row for relation "airport" violates check constraint "ck_airport_name_valid"
Подробности: Failing row contains (101, Short, Kazakhstan, Test, Test, 2026-10-12 19:05:06.953888, 2026-10-12 19:05:06.953888).
Services
Output
Result 78
214 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
124:1 (241 chars, 3 line breaks) LF UTF-8 4 spaces
Non-commercial use

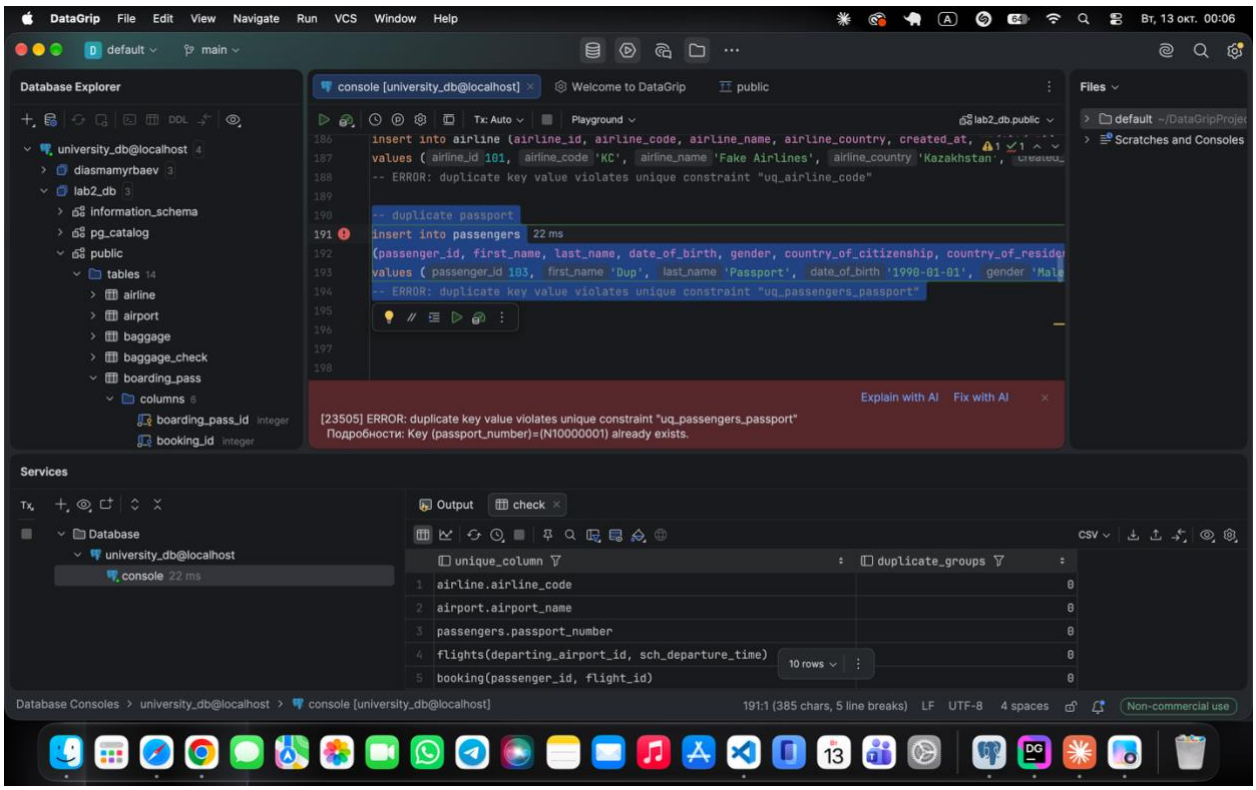


5. Add UNIQUE constraint to some columns in each table in database.

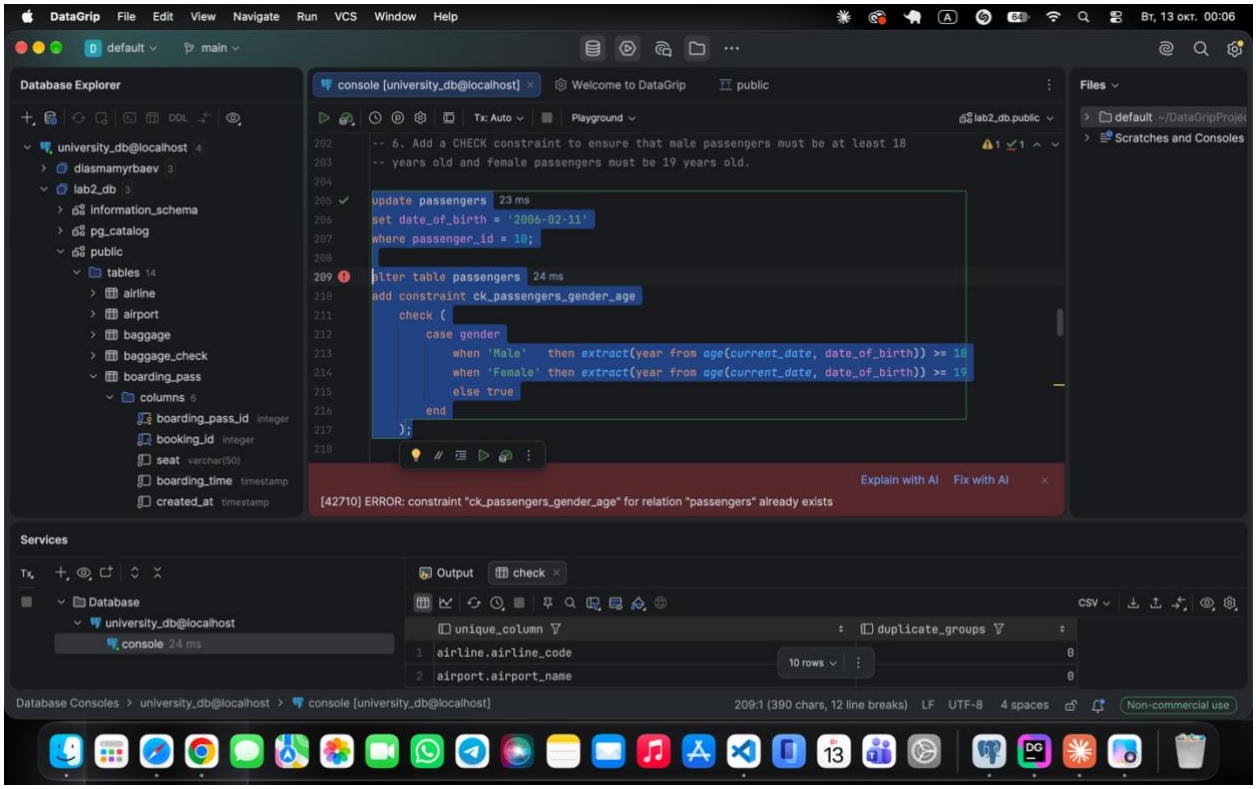


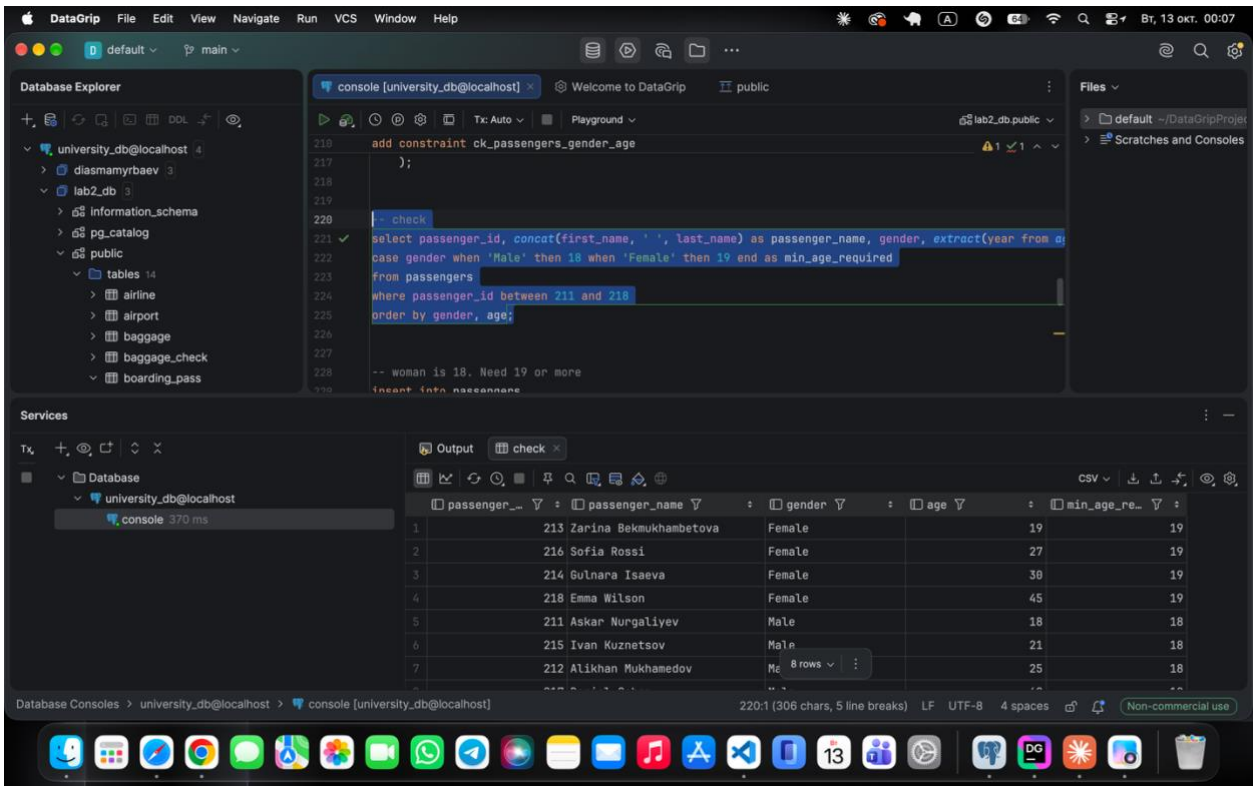

DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:05
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables 14
airline
airport
baggage
-- check
select 'airline.airline_code' as unique_column, count(*) as duplicate_groups 372 ms
from (select airline_code from airline group by airline_code having count(*) > 1) t
union all
select unique_column 'airport.airport_name', duplicate_groups count(*)
from (select airport_name from airport group by airport_name having count(*) > 1) t
union all
select unique_column 'passengers.passport_number', duplicate_groups count(*)
from (select passport_number from passengers group by passport_number having count(*) > 1) t
union all
Services
Output
check
Database
university_db@localhost
console 372 ms
unique_column
duplicate_groups
1 airline.airline_code 0
2 airport.airport_name 0
3 passengers.passport_number 0
4 flights(departing_airport_id, sch_departure_time) 0
5 booking(passenger_id, flight_id) 0
6 booking_flight(booking_id, flight_id) 0
7 baggage.booking_id 0
8 baggage_check.booking_id 0
9 boarding_pass.booking_id 0
10 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
153:1 (1518 chars, 30 line breaks) LF UTF-8 4 spaces
Non-commercial use

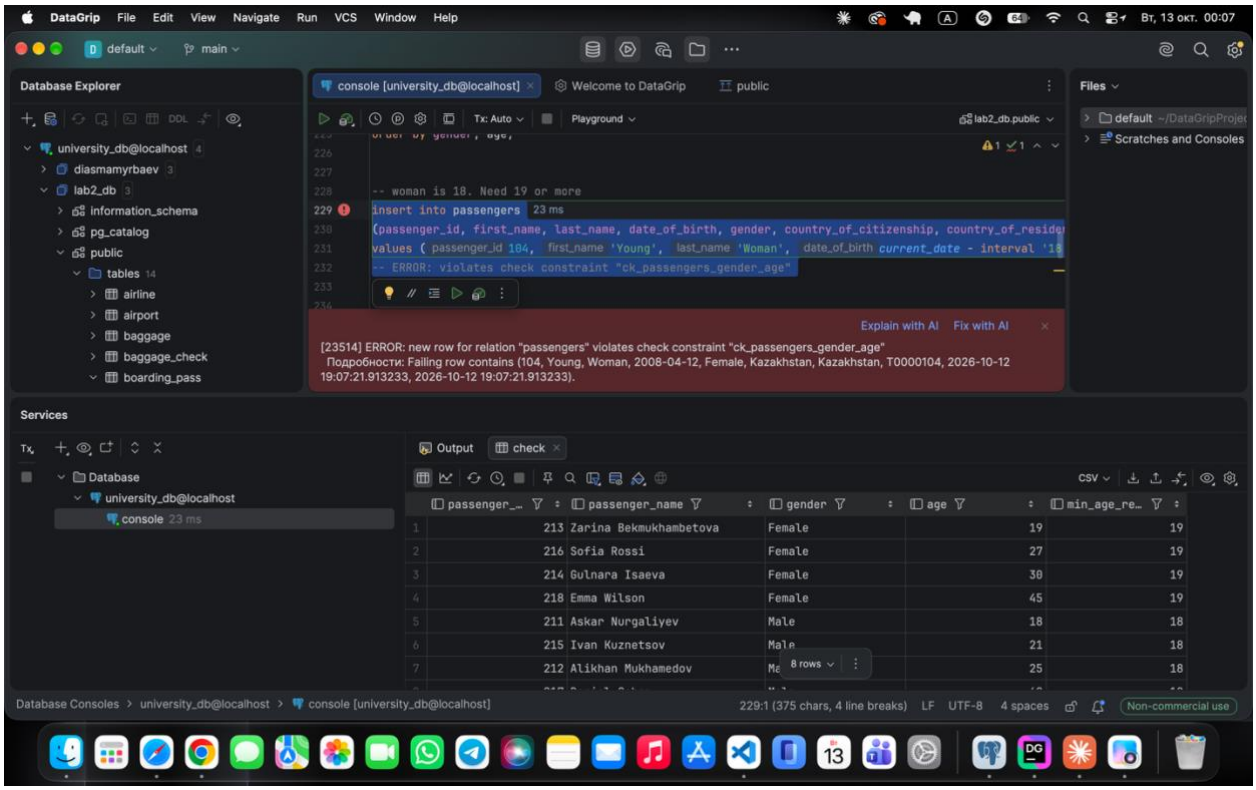

DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:06
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
from (select passenger_id, created_at from security_check group by 1, 2 having count(*) >
-- duplicate airport_code
insert into airline (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
values (airline_id 101, airline_code 'KC', airline_name 'Fake Airlines', airline_country 'Kazakhstan', created_
-- ERROR: duplicate key value violates unique constraint "uq_airline_code"
-- duplicate passport
insert into passengers
(passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_reside
values (passenger_id 103, first_name 'Dup', last_name 'Passport', date_of_birth '1990-01-01', gender 'Male
-- ERROR: duplicate key value violates unique constraint "uq_passengers_passport"
Explain with AI
Fix with AI
[23505] ERROR: duplicate key value violates unique constraint "uq_airline_code"
Подробности: Key (airline_code)=(KC) already exists.
boarding_pass
columns 8
boarding_pass_id integer
booking_id integer
Services
Output
check
console 18 ms
unique_column
duplicate_groups
1 airline.airline_code 0
2 airport.airport_name 0
3 passengers.passport_number 0
4 flights(departing_airport_id, sch_departure_time) 0
5 booking(passenger_id, flight_id) 0
10 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
186:1 (268 chars, 4 line breaks) LF UTF-8 4 spaces
Non-commercial use



6. Add a CHECK constraint to ensure that male passengers must be at least 18 years old and female passengers must be 19 years old.

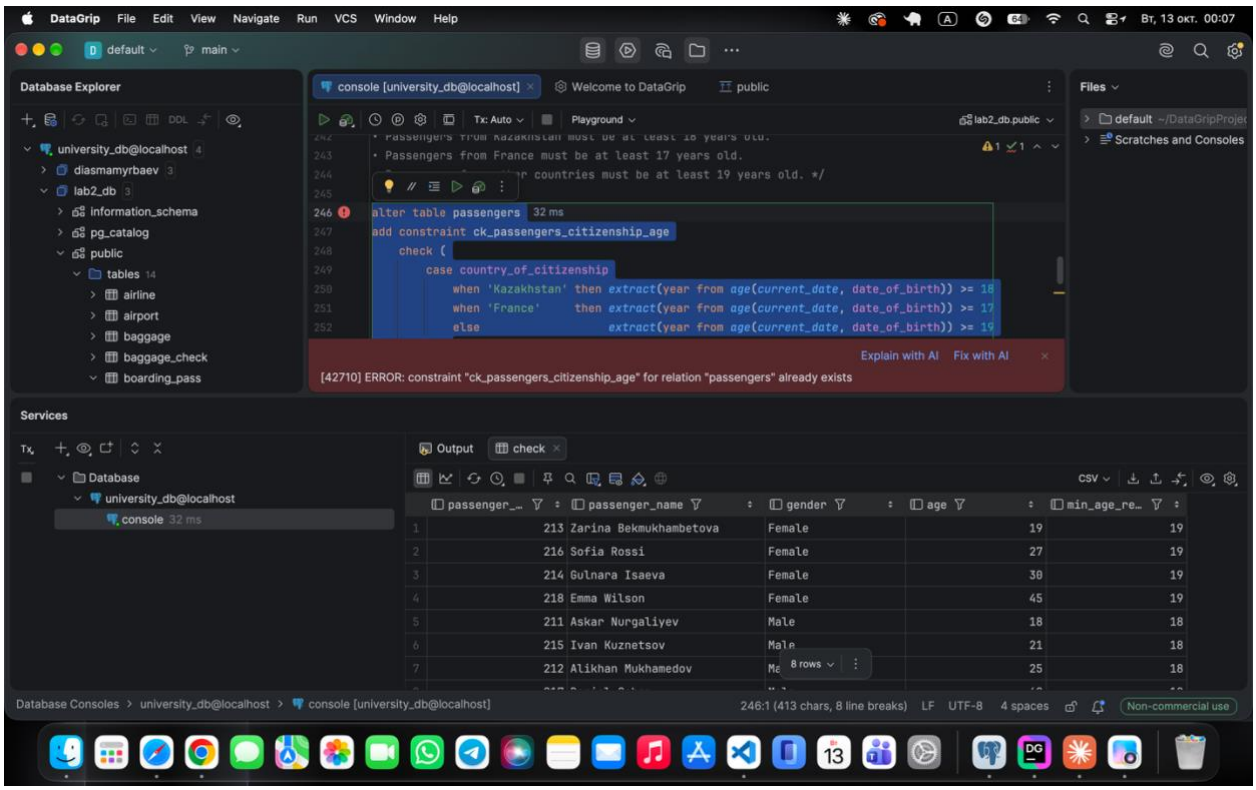


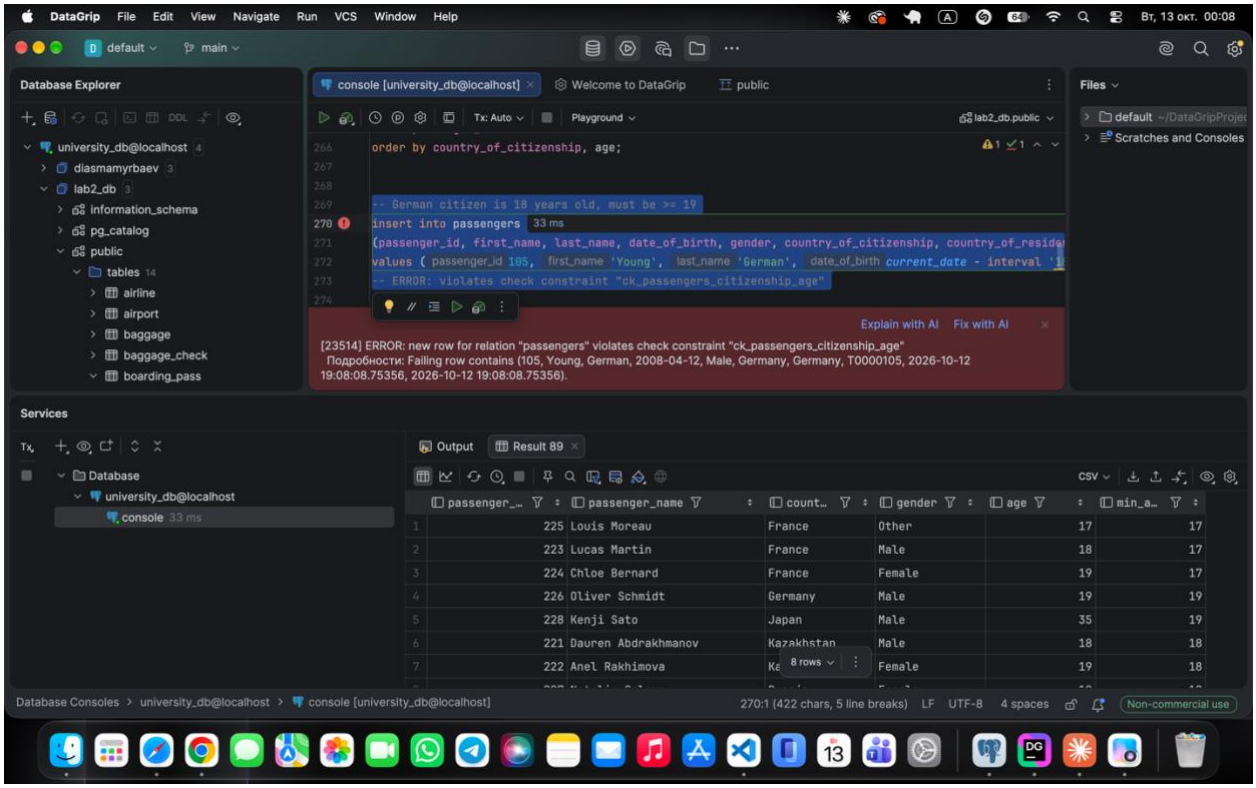


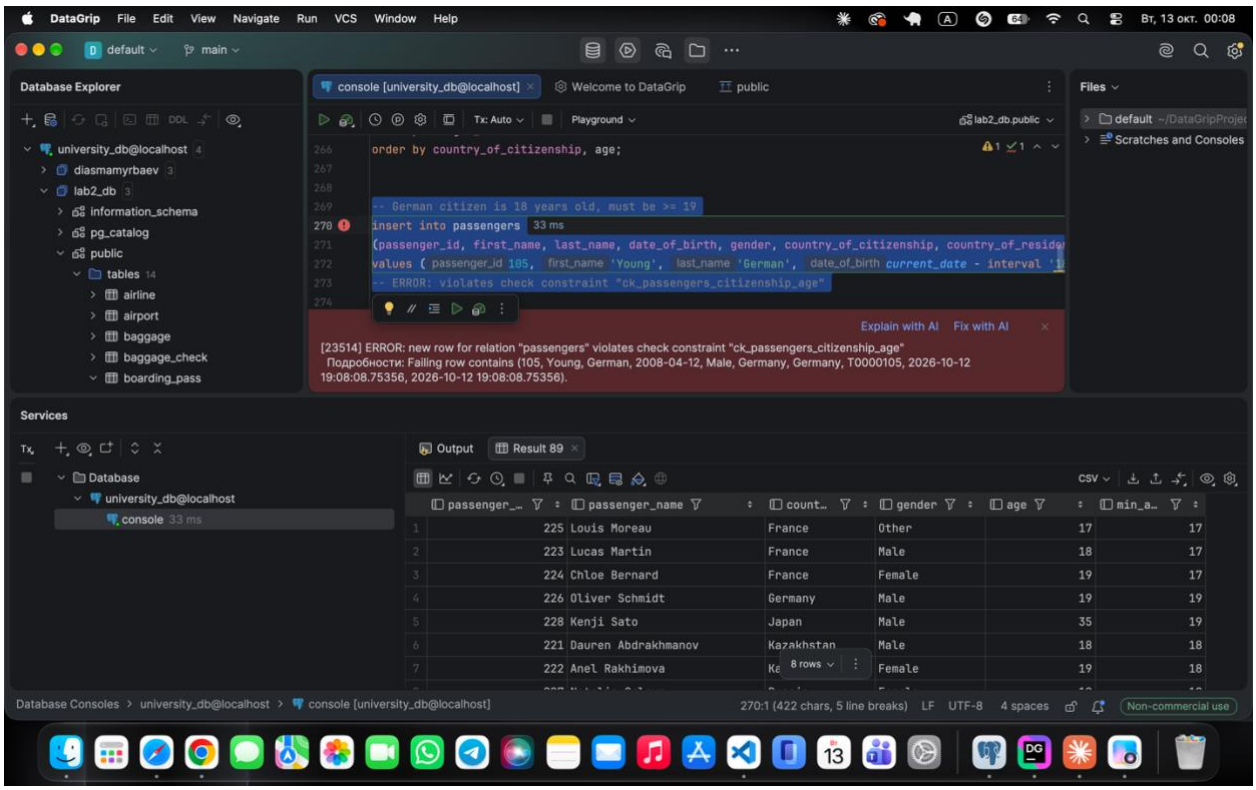


7. Add a CHECK constraint to add rule as follow (use column country_of_citizenship):

   - Passengers from Kazakhstan must be at least 18 years old.
   - Passengers from France must be at least 17 years old.
   - Passengers from other countries must be at least 19 years old.

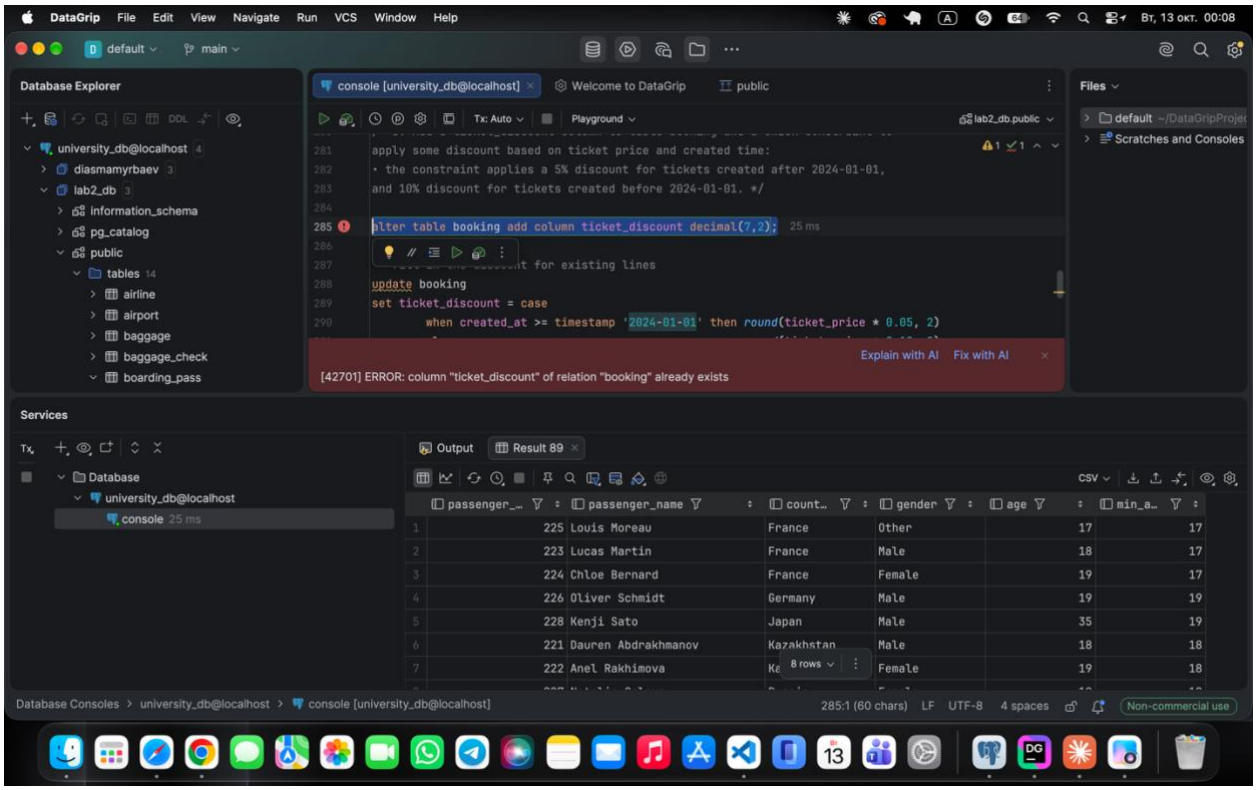

DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:07
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables 14
airline
airport
baggage
baggage_check
boarding_pass
• Passengers from France must be at least 17 years old.
alter table passengers 32 ms
add constraint ck_passengers_citizenship_age
check (
case country_of_citizenship
when 'Kazakhstan' then extract(year from age(current_date, date_of_birth)) >= 18
when 'France' then extract(year from age(current_date, date_of_birth)) >= 17
else extract(year from age(current_date, date_of_birth)) >= 19
Explain with AI
Fix with AI
[42710] ERROR: constraint "ck_passengers_citizenship_age" for relation "passengers" already exists
Services
Output
check
passenger_... passenger_name gender age min_age_re...
1 213 Zarina Bekmukhambetova Female 19 19
2 216 Sofia Rossi Female 27 19
3 214 Gulnara Isaeva Female 30 19
4 218 Emma Wilson Female 45 19
5 211 Askar Nurgaliyev Male 18 18
6 215 Ivan Kuznetsov Male 21 18
7 212 Alikhan Mukhamedov 25 18
8 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
246:1 (413 chars, 8 line breaks) LF UTF-8 4 spaces
Non-commercial use

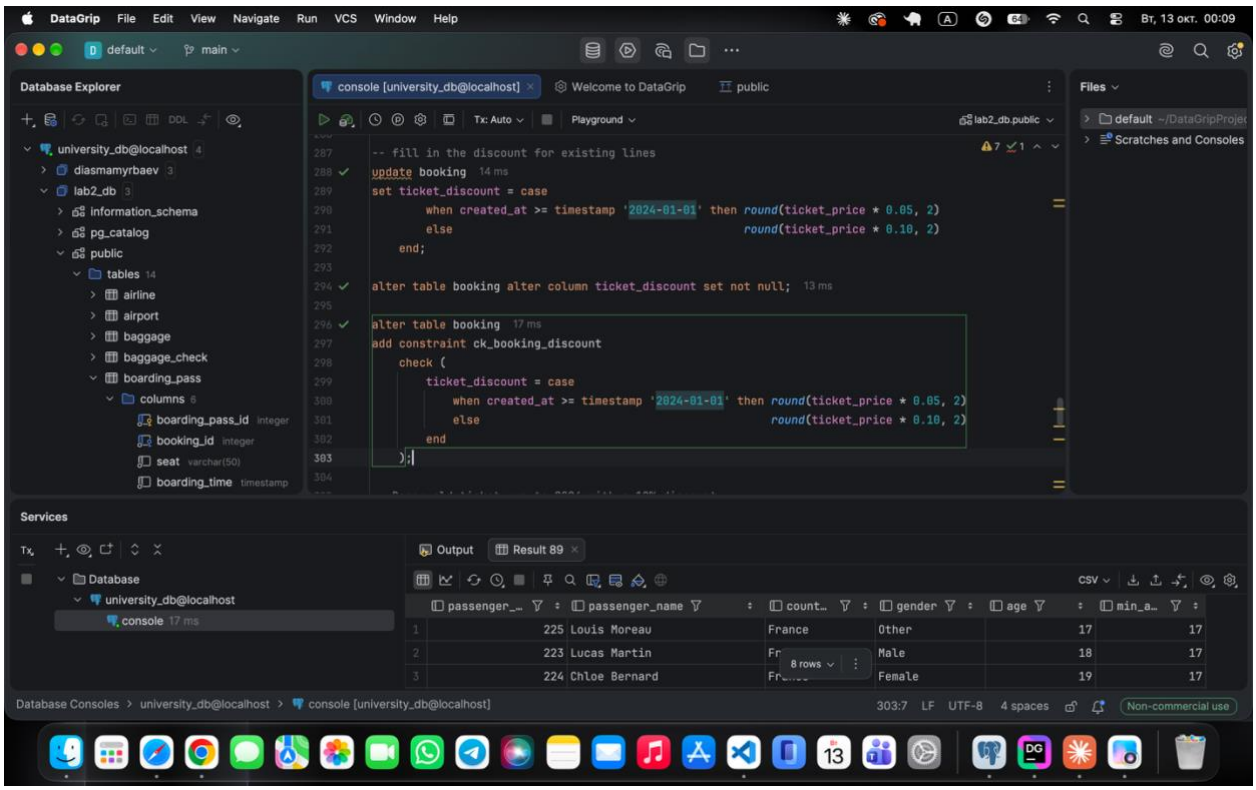

DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:08
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
order by country_of_citizenship, age;
-- German citizen is 18 years old, must be >= 19
insert into passengers 33 ms
(passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_reside
values (passenger_id 105, first_name 'Young', last_name 'German', date_of_birth current_date - interval '1
-- ERROR: violates check constraint "ck_passengers_citizenship_age"
Explain with AI
Fix with AI
[23514] ERROR: new row for relation "passengers" violates check constraint "ck_passengers_citizenship_age"
Подробности: Failing row contains (105, Young, German, 2008-04-12, Male, Germany, Germany, T0000105, 2026-10-12 19:08:08.75356, 2026-10-12 19:08:08.75356).
Services
Output
Result 89
passenger_... passenger_name count... gender age min_a...
1 225 Louis Moreau France Other 17 17
2 223 Lucas Martin France Male 18 17
3 224 Chloe Bernard France Female 19 17
4 226 Oliver Schmidt Germany Male 19 19
5 228 Kenji Sato Japan Male 35 19
6 221 Dauren Abdrakhmanov Kazakhstan Male 18 18
7 222 Anel Rakhimova Female 19 18
8 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
270:1 (422 chars, 5 line breaks) LF UTF-8 4 spaces
Non-commercial use

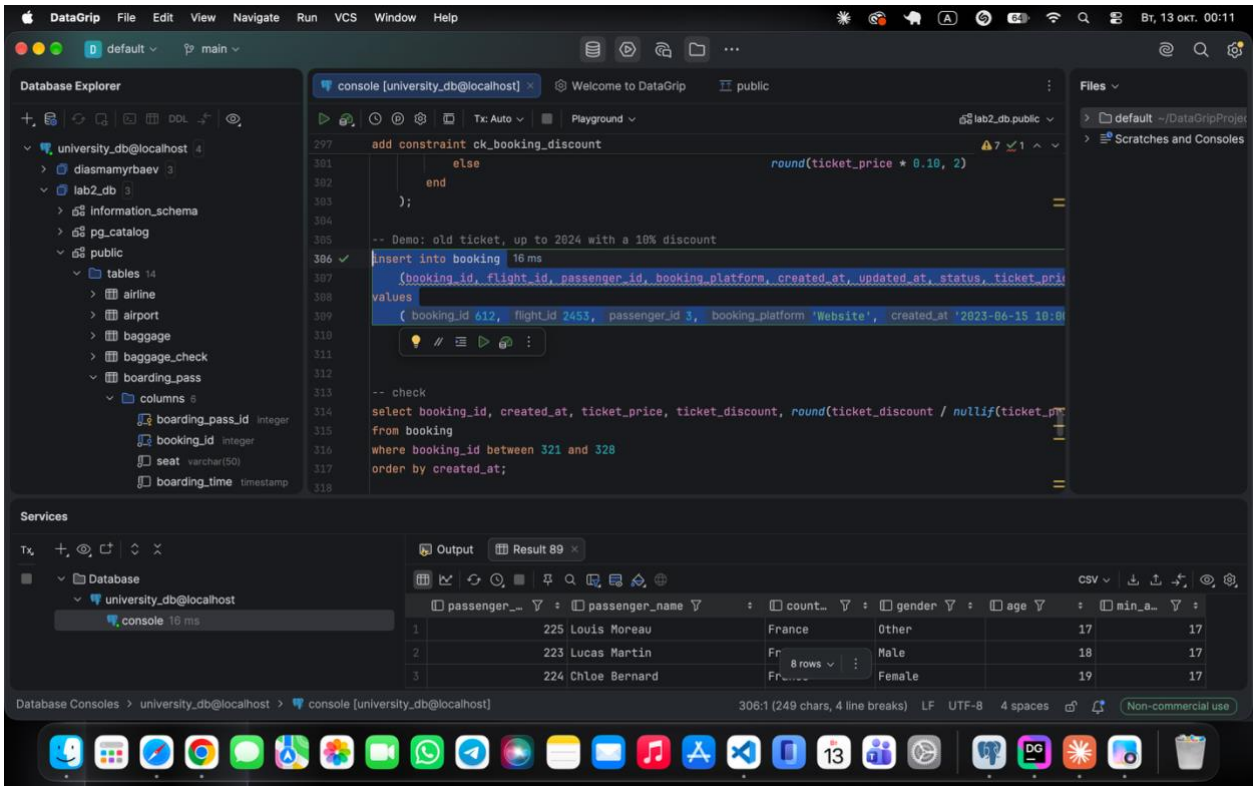


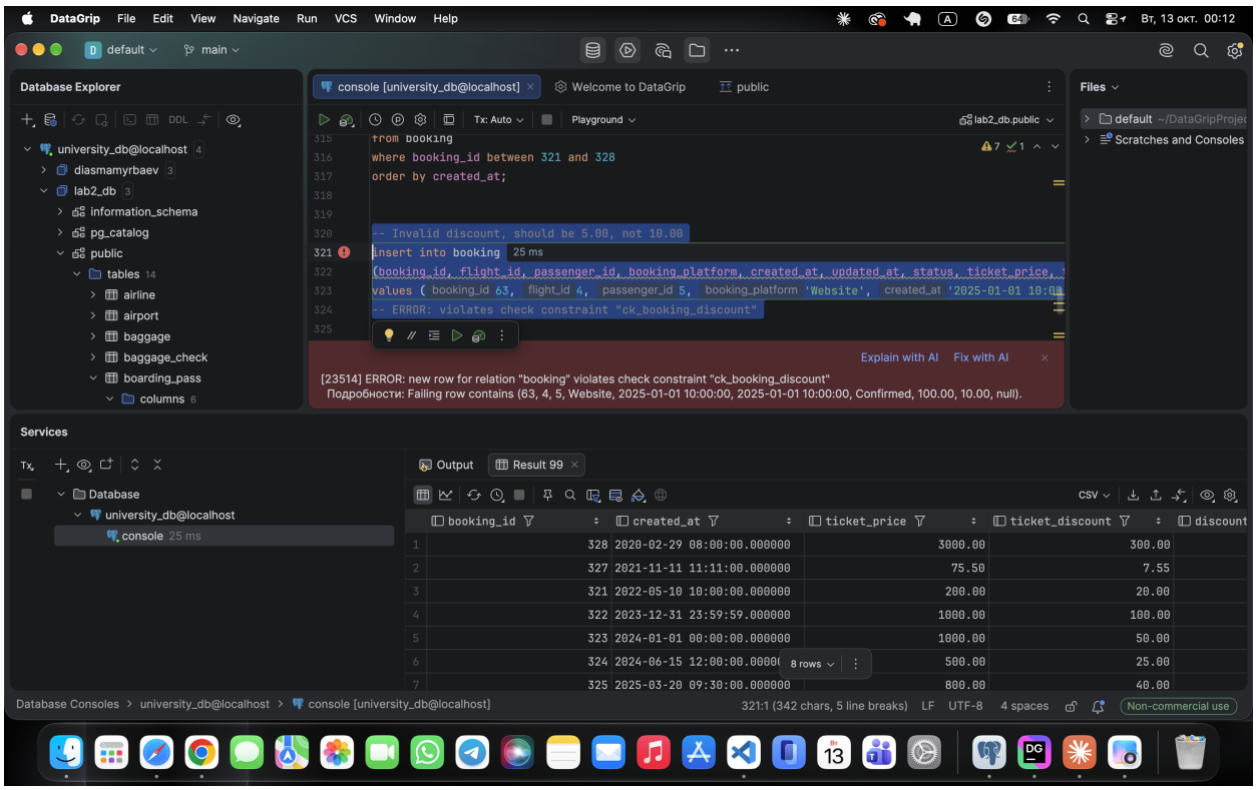
8. Add a ticket_discount column to table booking and a CHECK constraint to apply some discount based on ticket price and created time:

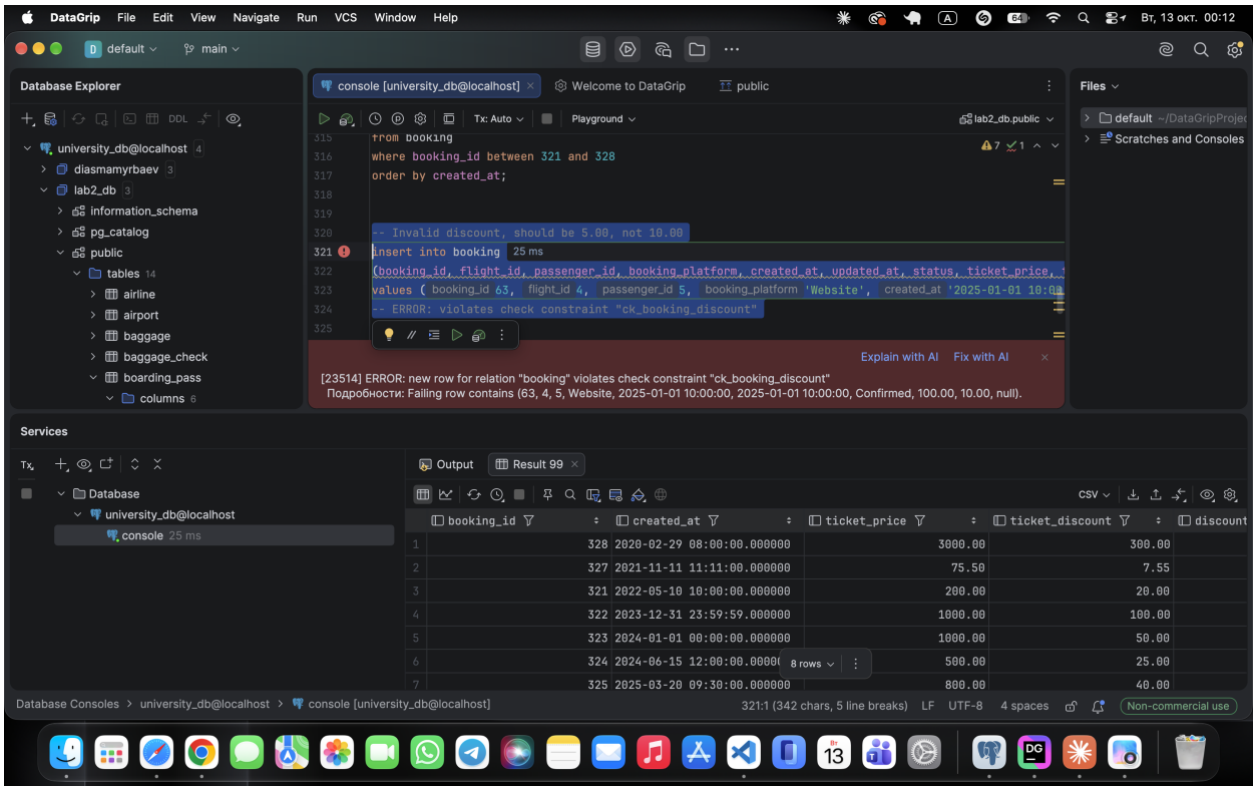
- the constraint applies a 5% discount for tickets created after 2024-01-01, and 10% discount for tickets created before 2024-01-01.



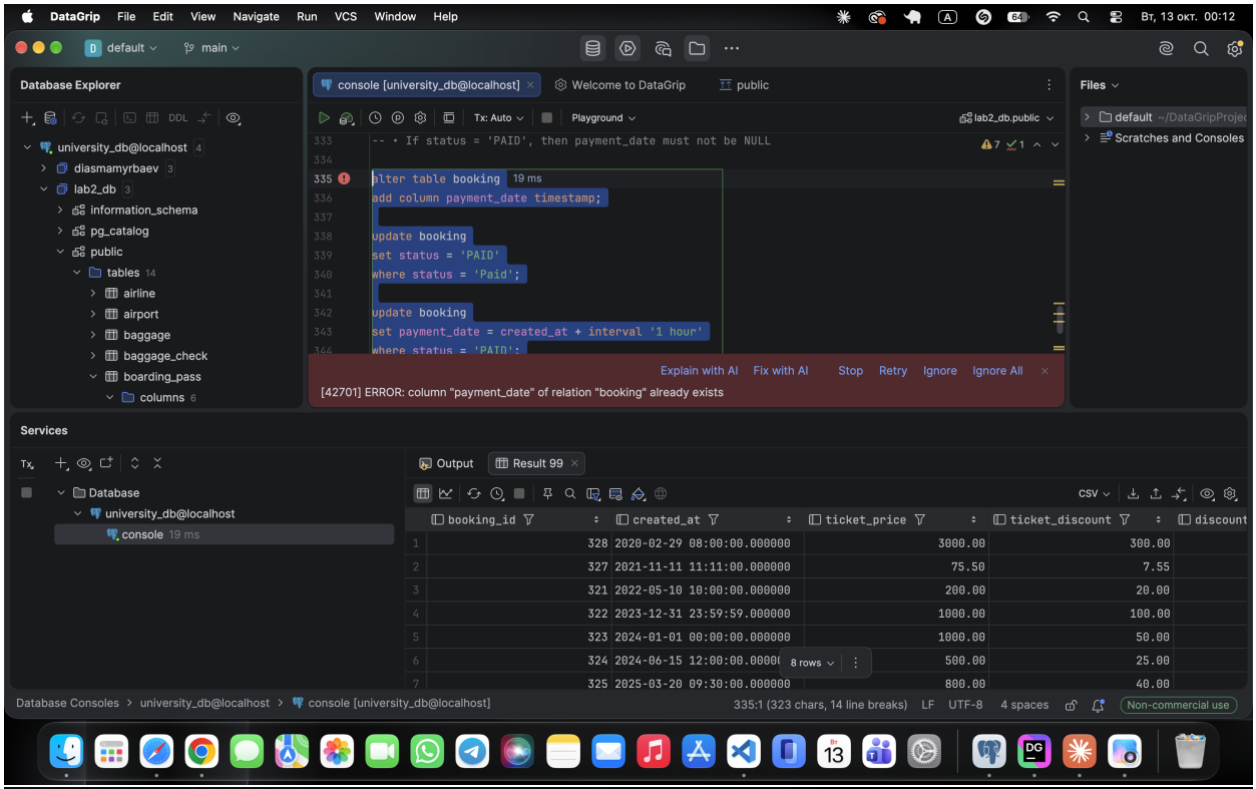

DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:09
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
-- fill in the discount for existing lines
update booking 14 ms
set ticket_discount = case
when created_at >= timestamp '2024-01-01' then round(ticket_price * 0.05, 2)
else round(ticket_price * 0.10, 2)
end;
alter table booking alter column ticket_discount set not null; 13 ms
alter table booking 17 ms
add constraint ck_booking_discount
check (
ticket_discount = case
when created_at >= timestamp '2024-01-01' then round(ticket_price * 0.05, 2)
else round(ticket_price * 0.10, 2)
end
);
Services
Output
Result 89
passenger_... passenger_name count... gender age min_a...
1 225 Louis Moreau France Other 17 17
2 223 Lucas Martin Fr 8 rows Male 18 17
3 224 Chloe Bernard Fr Female 19 17
Database Consoles > university_db@localhost > console [university_db@localhost]
303:7 LF UTF-8 4 spaces Non-commercial use

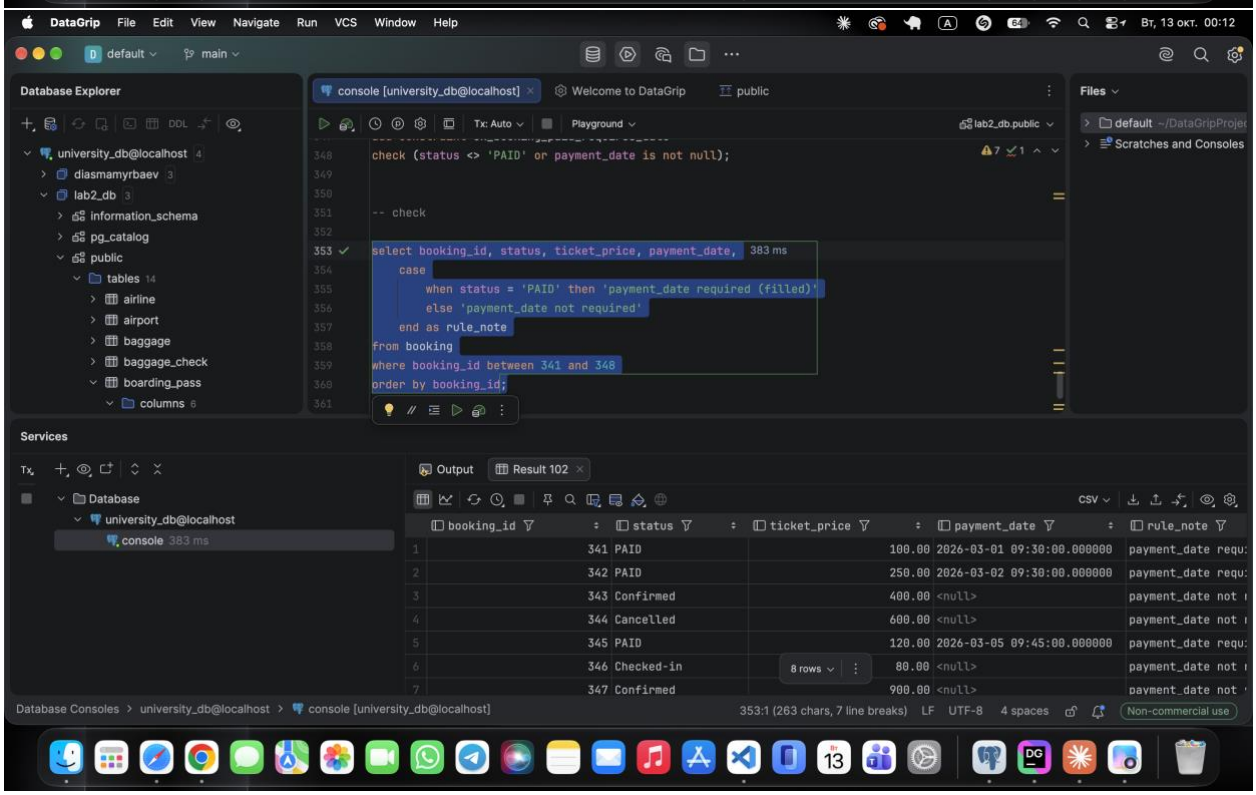

DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:11
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
add constraint ck_booking_discount
else round(ticket_price * 0.10, 2)
end
);
-- Demo: old ticket, up to 2024 with a 10% discount
insert into booking 16 ms
(booking_id, flight_id, passenger_id, booking_platform, created_at, updated_at, status, ticket_pri
values
(booking_id 612, flight_id 2453, passenger_id 3, booking_platform 'Website', created_at '2023-06-15 10:0
-- check
select booking_id, created_at, ticket_price, ticket_discount, round(ticket_discount / nullif(ticket_pr
from booking
where booking_id between 321 and 328
order by created_at;
Services
Output
Result 89
passenger_... passenger_name count... gender age min_a...
1 225 Louis Moreau France Other 17 17
2 223 Lucas Martin Fr 8 rows Male 18 17
3 224 Chloe Bernard Fr Female 19 17
Database Consoles > university_db@localhost > console [university_db@localhost]
306:1 (249 chars, 4 line breaks) LF UTF-8 4 spaces Non-commercial use

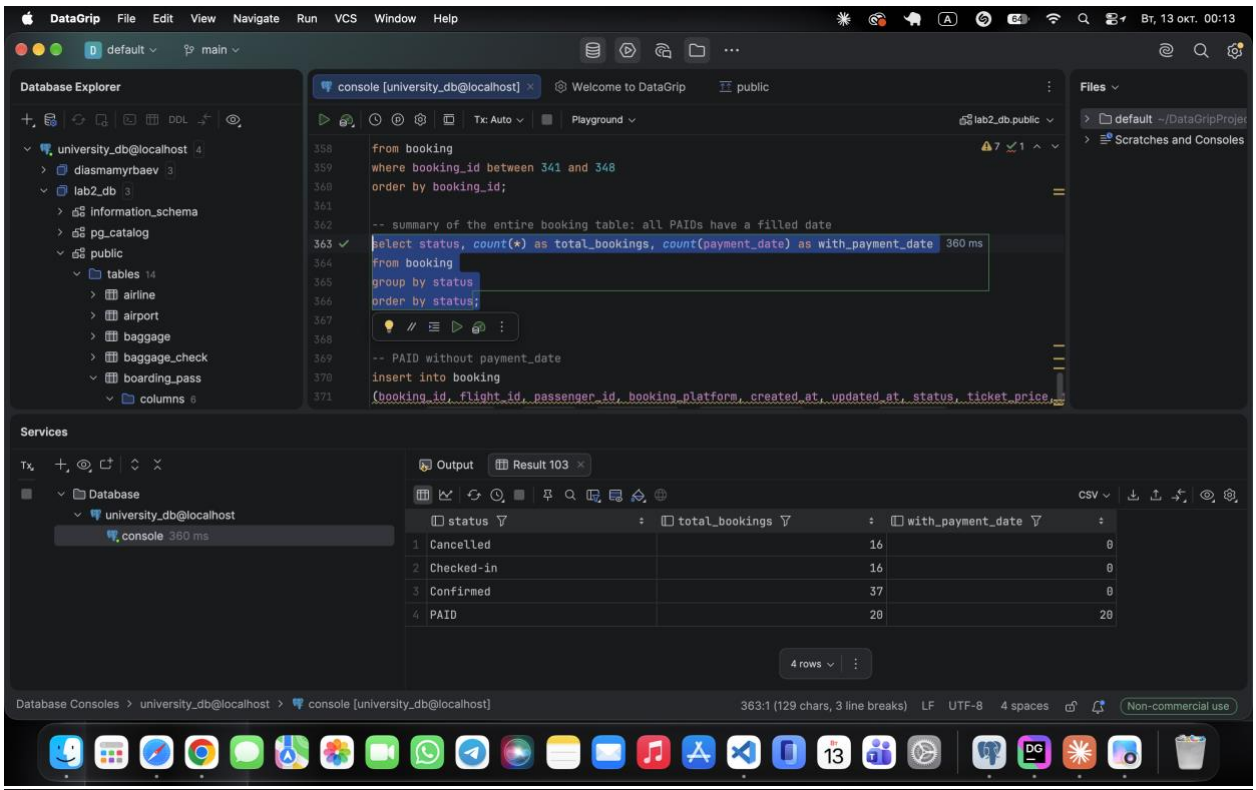


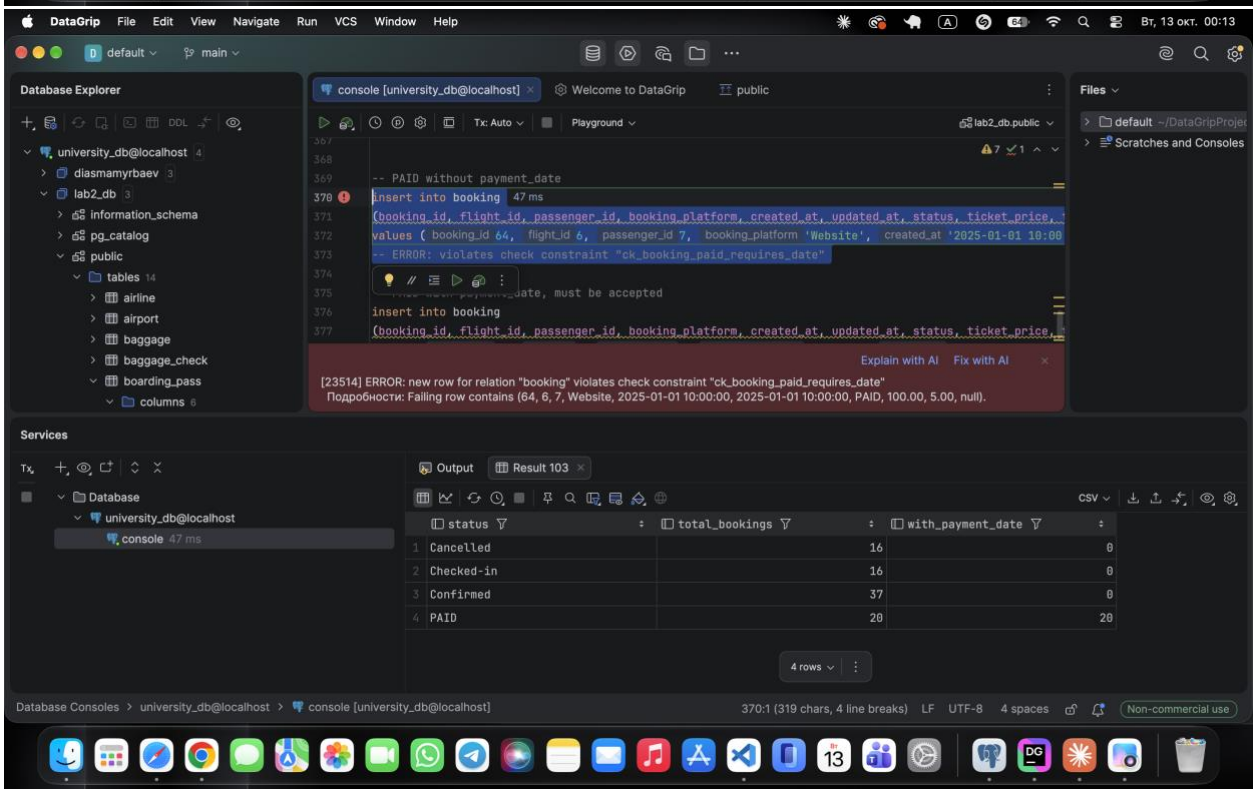


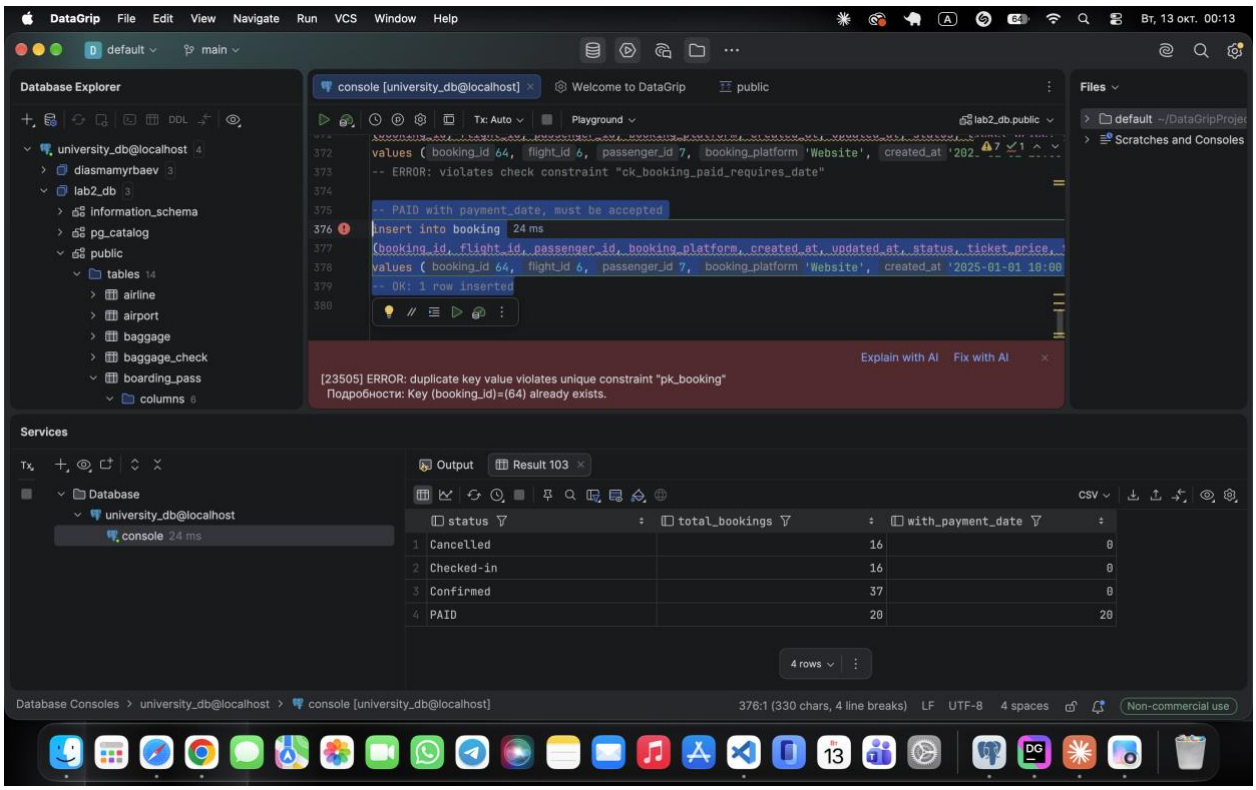
9. Add a CHECK constraint ensuring that the booking follows these rules:
   - If status = 'PAID', then payment_date must not be NULL


DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:12
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
-- * If status = 'PAID', then payment_date must not be NULL
alter table booking 19 ms
add column payment_date timestamp;
update booking
set status = 'PAID'
where status = 'Paid';
update booking
set payment_date = created_at + interval '1 hour'
Explain with AI Fix with AI Stop Retry Ignore Ignore All
[42701] ERROR: column "payment_date" of relation "booking" already exists
Services
Output
Result 99
booking_id created_at ticket_price ticket_discount discount
328 2020-02-29 08:00:00.000000 3000.00 300.00
327 2021-11-11 11:11:00.000000 75.50 7.55
321 2022-05-10 10:00:00.000000 200.00 20.00
322 2023-12-31 23:59:59.000000 1000.00 100.00
323 2024-01-01 00:00:00.000000 1000.00 50.00
324 2024-06-15 12:00:00.0000 500.00 25.00
325 2025-03-20 09:30:00.000000 800.00 40.00
8 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
335:1 (323 chars, 14 line breaks) LF UTF-8 4 spaces Non-commercial use


DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:12
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
check (status <> 'PAID' or payment_date is not null);
-- check
select booking_id, status, ticket_price, payment_date, 383 ms
case
when status = 'PAID' then 'payment_date required (filled)'
else 'payment_date not required'
end as rule_note
from booking
where booking_id between 341 and 348
order by booking_id;
Services
Output
Result 102
booking_id status ticket_price payment_date rule_note
341 PAID 100.00 2026-03-01 09:30:00.000000 payment_date requ
342 PAID 250.00 2026-03-02 09:30:00.000000 payment_date requ
343 Confirmed 400.00 <null> payment_date not
344 Cancelled 600.00 <null> payment_date not
345 PAID 120.00 2026-03-05 09:45:00.000000 payment_date requ
346 Checked-in 80.00 <null> payment_date not
347 Confirmed 900.00 <null> payment_date not
8 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
353:1 (263 chars, 7 line breaks) LF UTF-8 4 spaces Non-commercial use


DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:13
default main
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables 14
airline
airport
baggage
baggage_check
boarding_pass
columns 6
Tx: Auto
Playground
lab2_db.public
default ~/DataGripProje
Scratches and Consoles
from booking
where booking_id between 341 and 348
order by booking_id;
-- summary of the entire booking table: all PAIDs have a filled date
select status, count(*) as total_bookings, count(payment_date) as with_payment_date 360 ms
from booking
group by status
order by status;
-- PAID without payment_date
insert into booking
(booking_id, flight_id, passenger_id, booking_platform, created_at, updated_at, status, ticket_price,
Services
Database
university_db@localhost
console 360 ms
Output
Result 103
CSV
status
total_bookings
with_payment_date
1 Cancelled 16 0
2 Checked-in 16 0
3 Confirmed 37 0
4 PAID 20 20
4 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
363:1 (129 chars, 3 line breaks) LF UTF-8 4 spaces
Non-commercial use


DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:13
default main
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables 14
airline
airport
baggage
baggage_check
boarding_pass
columns 6
Tx: Auto
Playground
lab2_db.public
default ~/DataGripProje
Scratches and Consoles
-- PAID without payment_date
insert into booking 47 ms
(booking_id, flight_id, passenger_id, booking_platform, created_at, updated_at, status, ticket_price,
values (booking_id 64, flight_id 6, passenger_id 7, booking_platform 'Website', created_at '2025-01-01 10:00
-- ERROR: violates check constraint "ck_booking_paid_requires_date"
date, must be accepted
insert into booking
(booking_id, flight_id, passenger_id, booking_platform, created_at, updated_at, status, ticket_price,
Explain with AI
Fix with AI
[23514] ERROR: new row for relation "booking" violates check constraint "ck_booking_paid_requires_date"
Подробности: Failing row contains (64, 6, 7, Website, 2025-01-01 10:00:00, 2025-01-01 10:00:00, PAID, 100.00, 5.00, null).
Services
Database
university_db@localhost
console 47 ms
Output
Result 103
CSV
status
total_bookings
with_payment_date
1 Cancelled 16 0
2 Checked-in 16 0
3 Confirmed 37 0
4 PAID 20 20
4 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
370:1 (319 chars, 4 line breaks) LF UTF-8 4 spaces
Non-commercial use


DataGrip File Edit View Navigate Run VCS Window Help
Вт, 13 окт. 00:13
default main
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Files
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables
airline
airport
baggage
baggage_check
boarding_pass
columns
Tx: Auto
Playground
lab2_db.public
values (booking_id 64, flight_id 6, passenger_id 7, booking_platform 'Website', created_at '202
-- ERROR: violates check constraint "ck_booking_paid_requires_date"
-- PAID with payment_date, must be accepted
insert into booking 24 ms
(booking_id, flight_id, passenger_id, booking_platform, created_at, updated_at, status, ticket_price,
values (booking_id 64, flight_id 6, passenger_id 7, booking_platform 'Website', created_at '2025-01-01 10:00
-- OK: 1 row inserted
Explain with AI
Fix with AI
[23505] ERROR: duplicate key value violates unique constraint "pk_booking"
Подробности: Key (booking_id)=(64) already exists.
Default ~/DataGripProje
Scratches and Consoles
Services
Database
university_db@localhost
console 24 ms
Output
Result 103
CSV
status
total_bookings
with_payment_date
1 Cancelled 16 0
2 Checked-in 16 0
3 Confirmed 37 0
4 PAID 20 20
4 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
376:1 (330 chars, 4 line breaks) LF UTF-8 4 spaces
Non-commercial use